Virtual Humanoid User Flow Template

AI Execution Specification for the Loan Underwriting AI Agent Dashboard

Includes detailed examples for all action types: navigation, UI/UX page maps, selectors, coordinates, pop-ups, drilldowns, application timelines, validation, recovery, and humanoid narration.

How to use this document: This template is written so an AI agent or virtual humanoid can read the flow, identify the target screen, locate the UI element, perform the required action, validate the result, narrate like a human, and recover if the UI does not behave as expected.

Document Field	Value
Prototype application name	Loan Underwriting AI Agent Dashboard
Primary user role	Credit Officer or business reviewer watching the humanoid perform the underwriting review
Execution actor	Virtual humanoid operating the loan dashboard in a browser
Created / regenerated on	2026-04-25
Template status	Filled production spec — AI Loan Agent demo flow

1. Prototype Scenario Used Throughout This Template

This document uses one consistent prototype so every table is self-explanatory. The example application is the Loan Underwriting AI Agent Dashboard — a browser-based tool that lets a credit officer review AI-generated loan decisions, inspect individual application timelines, and perform live human-in-the-loop (HIL) actions.

Area	Prototype Example
Web application	Loan Underwriting AI Agent Dashboard
Purpose	A browser-based dashboard that lets a virtual humanoid open the desktop snapshot, navigate to the overview, drill into individual application timelines for three contrasting decisions, and perform a live HIL approval action — all while narrating the AI agent reasoning to a human observer.
Main desktop URL	http://134.33.132.134/maria-aegis-fe
Viewport baseline	1440 x 900 browser viewport, 100 percent zoom
Coordinate origin	Top-left of browser content area after the address bar is excluded
Primary business flow	Review today's loan batch: read the queue snapshot on the desktop, explain the overview dashboard KPIs and charts, walk through three application timelines (MEHTA VARUN — clean STP approval, PILLAI SAVITA — hard DTI rejection, KRISHNAN DEEPAK — HIL escalation), navigate to the HIL kanban, open the KRISHNAN DEEPAK modal, and perform a live approval decision.
Primary audience	A human observer watching the humanoid operate the dashboard and explain each loan decision.

2. Complete Flow Identification Template

Field	Required	Description	Filled Example
flow_id	Yes	Unique ID for the complete business journey.	FLOW-LOAN-UNDERWRITING-DEMO-001
flow_name	Yes	Human readable name.	AI Loan Agent — End to End Underwriting Demo
flow_version	Yes	Version used by the AI runtime.	1.0
parent_flow_id	No	Used when this flow is a child flow.	None
business_goal	Yes	Why the user wants the flow performed.	Demonstrate how the AI agent processes loans: auto-approves clean cases, rejects hard policy breaches, and escalates genuinely ambiguous cases to human review — with full step-by-step reasoning visible.
persona	Yes	Role being simulated by humanoid.	Credit Officer reviewing the daily loan underwriting batch
trigger_utterance	Yes	User command that starts the flow.	Show me today's loan pipeline and walk me through three decisions — one approval, one rejection, and one that needs my review.
entry_screen_id	Yes	First screen where execution begins.	SCR-001-DESKTOP
expected_end_screen_id	Yes	Final screen after successful execution.	SCR-006-HIL-MODAL
narration_mode	Yes	How much the humanoid should speak.	Detailed guided walkthrough — explain every screen, every metric, and every agent decision step
evidence_required	Yes	Whether screenshots/logs are required.	Yes, capture screenshot at each major screen transition and after the HIL approval action

3. Required Runtime Parameters

Parameter	Required	Type	Example	How AI Uses It
session_id	Yes	String	LOAN-AGENT-DEMO-001	Correlates all actions, screenshots, and log entries for this execution run.
user_role	Yes	String	Credit Officer	Determines narration tone and explanation depth. Credit Officer gets full technical detail on CIBIL, DTI, and policy flags.
base_url	Yes	String	http://134.33.132.134/maria-aegis-fe	Root URL prepended to every route. All navigation is relative to this base.
viewport_width	Yes	Number	1440	Used to calculate element coordinates and validate region bounds throughout the flow.
viewport_height	Yes	Number	900	Used to calculate element coordinates and validate region bounds throughout the flow.
speaker_style	Yes	Enum	Professional, explanatory	Controls humanoid narration tone. Options: Professional, concise, explanatory. Use explanatory for this demo.
application_1_id	Yes	String	APP-2026-00001	Application ID for MEHTA VARUN — the clean STP approval case. Used to navigate to /applications/APP-2026-00001.
application_1_name	Yes	String	MEHTA VARUN	Applicant name used to locate the correct row in the applications table before clicking Open page.
application_2_id	Yes	String	APP-2026-00003	Application ID for PILLAI SAVITA — the hard DTI rejection case. Used to navigate to /applications/APP-2026-00003.
application_2_name	Yes	String	PILLAI SAVITA	Applicant name used to locate the correct row in the applications table before clicking Open page.
application_3_id	Yes	String	APP-2026-00012	Application ID for KRISHNAN DEEPAK — the HIL escalation case. Used to locate the kanban card in the Awaiting Review lane.
application_3_name	Yes	String	KRISHNAN DEEPAK	Applicant name used to locate the card in the HIL Awaiting Review lane and validate the modal header after opening.
hil_action	Yes	Enum	approve	The live decision the humanoid performs for KRISHNAN DEEPAK inside the HIL modal. Set by the operator before the flow runs. Valid values: approve, reject, modify_approve.

4. UI/UX Screen Inventory: Where the Humanoid Goes

The humanoid visits 6 distinct screens during the complete demo flow. Below is the inventory of each screen, its purpose, primary content regions, and expected visible evidence.

Screen ID	Screen / Page Name	URL or Route	Purpose in Flow	Primary Content Regions	Expected Visible Evidence
SCR-001-DESKTOP	Desktop Home	/	Act 1: System orientation. Humanoid explains what the AI agent dashboard does and the loan underwriting pipeline at a high level.	Center hero card, left queue snapshot panel, right decision mix panel, bottom taskbar	Queue Snapshot (Need Review, Reviewed, Accepted, Rejected tiles), Decision Mix bars, three taskbar icons, AI Loan Agent title
SCR-002-OVERVIEW	Overview Dashboard	/overview	Act 2: Dashboard walkthrough. Humanoid explains KPIs, charts, pipeline stages, batch volume, rejection reasons, and the flow of data through the system.	KPI strip (5 cards), three doughnut charts, two bar charts, 7-step pipeline row, four metric cards bottom	14 Applications Today, 57% STP Approved, HIL Queue 5, Decision/Risk/Employer distributions, Daily Batch Volume, Top Rejection Reasons, pipeline steps 01-07, STP Rate/Accuracy/False Reject/LMS metrics
SCR-005-HIL-REVIEW	HIL Review Kanban	/hil-review	Act 3-4: Humanoid reads the four kanban lanes, explains each one, then opens KRISHNAN DEEPAK card from Awaiting Review lane.	Outer gate card header, four lane columns (Awaiting Review / Under Review / CO Approved / CO Rejected), kanban cards per lane	Four lane headers with counts (1/2/0/5), KRISHNAN DEEPAK card visible in Awaiting Review lane with score 75 and Low Risk chip
SCR-006-HIL-MODAL	HIL Application Modal	Modal overlay on /hil-review	Act 4: Deep dive into KRISHNAN DEEPAK case. Humanoid scrolls and narrates the full audit timeline, reads key metrics, explains the dilemma, thinks out loud about the decision, then clicks [+] Approve button.	Modal header (title, close button), left audit trail panel (EVT_DEDUP, EVT_UNKNOWN, EVT_FEATURES steps), right borrower profile (name, HIL PENDING status, risk score 75/100), Key Metrics grid (CIBIL 789, DTI 19.45%, Bounce 0/mo, Spending Ratio 0.53), footer	KRISHNAN DEEPAK name, APP-2026-00012, HIL PENDING label, score 75, all timeline steps visible with expand capability, all Key Metrics visible, four decision buttons at bottom, modal confirms approval after button click

Screen ID	Screen / Page Name	URL or Route	Purpose in Flow	Primary Content Regions	Expected Visible Evidence
				action buttons (Cancel, Modify & Approve, Reject, Approve)	
SCR-003-APPLICATIONS	All Applications	/applications	Act 5-8: After user interrupts and asks for recent applications. Humanoid navigates here and searches for three specific applicants (MEHTA VARUN approved, KRISHNAN DEEPAK pending, PILLAI SAVITA rejected) and clicks Open page for each.	Top filter bar (search, status filter, risk filter, export), three mini-charts (Decision/Risk/Confidence distributions), application register table with 14 rows	14 rows showing all applications, MEHTA VARUN visible (APP-2026-00001, Conditional Approval, Low Risk, 100% confidence), PILLAI SAVITA visible (APP-2026-00003, Rejected, High Risk), KRISHNAN DEEPAK visible (APP-2026-00012, HIL Pending, Medium Risk), Open page links in Route column
SCR-004-APP-DETAIL	Application Detail	/applications/:id	Act 6-8: Humanoid opens three different application detail pages one-by-one (MEHTA VARUN happy path, PILLAI SAVITA rejection, KRISHNAN DEEPAK pending). For each: reads the summary card, then scrolls the full Agent Timeline and narrates every step from KYC_VALIDATION through FINAL_DECISION.	Header card (back button, applicant name, app ID, reasoning summary), Application Summary grid (Employer, Income, CIBIL, DTI, Risk, Decision, Approved Amount, Confidence), Agent Timeline section with TimelineStep nodes (one per pipeline stage, data-driven)	Applicant name in header, app ID (APP-2026-00001/00003/00012), full timeline visible and scrollable (8 steps for clean approvals, 9 steps for escalated/rejected cases), every step shows status dot, tool name, step name, reasoning text, and output payload. FINAL_DECISION step output_data contains the terminal outcome.

After completing all three application detail walkthrough narratives, the humanoid returns to explain the credit officer's final authority and delivers the closing narration (Act 9).

5. Screen Layout Region Map

Use this to tell the AI where to look before using a selector or coordinate. All coordinates are based on a 1440 × 900 viewport. Origin (0,0) is top-left of the content area, excluding the browser chrome.

Region ID	Applies To	Pixel Bounds x1,y1,x2,y2	Description	Example AI Instruction
REG-TASKBAR	All screens	520,700,940,780	Bottom dock bar with three AppIcon links (Overview, HIL Review, Applications) and live clock.	Use taskbar to navigate between Overview, HIL Review, and Applications screens. Click the appropriate icon.
REG-SNAPSHOT-LEFT	SCR-001-DESKTOP	20,30,380,240	Left panel: Today's Queue Snapshot with four metric tiles (Need Review, Reviewed, Accepted, Rejected).	Read the four tiles to understand today's queue status before navigating to detailed screens.
REG-DECISION-MIX	SCR-001-DESKTOP	1150,30,1490,230	Right panel: Decision Mix showing Accepted %, Rejected %, and Under Review % as progress bars.	Read the decision mix percentages to see the overall approval/rejection balance.
REG-HERO-CENTER	SCR-001-DESKTOP	300,240,1140,470	Center Mac-window hero card with title 'AI Loan Agent' and subtitle 'Virtual Credit Underwriter'.	Confirm desktop has loaded by detecting the AI Loan Agent hero card and title text.
REG-KPI-STRIP	SCR-002-OVERVIEW	0,0,1440,90	Five KPI cards across the top: Applications Today, STP Approved %, HIL Queue, Rejected, Avg TAT.	Read each KPI card value. These are live values from today's batch. Treat numbers as examples not strict assertions.
REG-CHARTS-ROW1	SCR-002-OVERVIEW	0,90,1440,340	Three doughnut charts: Decision Distribution (Approved/Rejected/HIL Pending), Risk Distribution (Low/High/Medium), Employer Split (Private/Government).	Explain each doughnut chart. Read the legend and segment percentages.
REG-CHARTS-ROW2	SCR-002-OVERVIEW	0,340,1440,540	Two bar charts: Daily Batch Volume (Mon-Today, Processed vs Rejected bars), Top Rejection Reasons (HIGH_DTI, LOW_CIBIL, HIGH_RISK,	Explain batch volume trend and which rejection reasons dominate. HIGH_DTI should be the longest bar.

Region ID	Applies To	Pixel Bounds x1,y1,x2,y2	Description	Example AI Instruction
			PROXY_SCORE horizontal bars).	
REG-PIPELINE	SCR-002-OVERVIEW	0,540,1440,640	Seven-step pipeline strip showing numbered stages: 01 Data Intake, 02 Feature Eng, 03 Rule Engine, 04 LLM Reasoning, 05 HIL Review, 06 Audit Trail, 07 LMS Push.	Read all seven pipeline stages in order. Explain what each stage does in the AI decision process.
REG-METRICS-BOTTOM	SCR-002-OVERVIEW	0,640,1440,820	Four metric cards at bottom: STP Rate (57%), Decision Accuracy (85%), False Reject Rate (18%), LMS Push Success (100%).	Read each metric and explain what it means. STP Rate = auto-approval rate, Accuracy = model correctness, False Reject = unnecessary rejections, LMS Push = successful delivery.
REG-FILTER-BAR	SCR-003-APPLICATIONS	600,100,1440,165	Top filter and search bar: search input (Search applicant or app id), All Status dropdown, All Risk Tiers dropdown, Export CSV button.	Use search to find applicants by name or app ID. Use filters to narrow by decision status or risk level.
REG-MINI-CHARTS	SCR-003-APPLICATIONS	0,220,1440,380	Three small charts above the table: Decision Distribution, Risk Distribution, Confidence Distribution.	These summarize the applications displayed in the table below. Confirm distribution matches what was seen in the main overview.
REG-APP-TABLE	SCR-003-APPLICATIONS	0,380,1440,1000	Application register table with 14 rows. Columns: Applicant, Loan Amount, Decision, Risk, Confidence, HIL, Created, Route (Open page link).	Locate applicants by name: MEHTA VARUN (approved), PILLAI SAVITA (rejected), KRISHNAN DEEPAK (pending). Click Open page link to navigate to detail.
REG-HIL-LANES	SCR-005-HIL-REVIEW	0,80,1440,850	Four kanban lanes in a horizontal row: Awaiting Review (1 card), Under Review (2 cards), CO Approved (0), CO Rejected (5 cards). Each lane has a header with count badge.	Read all four lane headers and counts. Locate KRISHNAN DEEPAK in Awaiting Review lane. Click the card to open the modal.
REG-KANBAN-CARD	SCR-005-HIL-REVIEW	varies by lane	Individual kanban card showing: app ID, applicant name, loan	Each card displays the applicant's key info. KRISHNAN DEEPAK card

Region ID	Applies To	Pixel Bounds x1,y1,x2,y2	Description	Example AI Instruction
			amount, tenure, risk chip (color-coded), risk score bar.	shows APP-2026-00012, Low Risk chip, score 75 bar.
REG-MODAL-HEADER	SCR-006-HIL-MODAL	30,10,1200,100	Modal header: title 'Application Review & Status', app ID (APP-2026-00012), and Close button (rose/pink color) top-right.	Confirm modal opened with correct applicant. Use Close button to dismiss modal if needed.
REG-MODAL-LEFT	SCR-006-HIL-MODAL	30,120,460,800	Left panel: scrollable audit timeline with event cards. Each event shows timestamp, code (EVT_DEDUP, EVT_UNKNOWN, EVT_FEATURES), step name, description, and code output snippet.	Scroll through all audit events. Read each event title and description. Expand events to see detailed outputs (KYC validation, document analysis, income verification, etc.).
REG-MODAL-RIGHT	SCR-006-HIL-MODAL	480,120,1200,800	Right panel: Applicant name and ID, HIL PENDING status label, risk score (75/100) with orange progress bar, KEY METRICS grid (4 cells: CIBIL Score, DTI Ratio, Bounce Frequency, Spending Ratio).	Read applicant name, confirm HIL PENDING status, read the score out of 100, and read all four key metrics (CIBIL 789, DTI 19.45%, Bounce 0/mo, Spending 0.53).
REG-MODAL-FOOTER	SCR-006-HIL-MODAL	30,820,1200,880	Footer row with four action buttons: Cancel (gray), [-] Modify & Approve (amber/orange), [x] Reject (red), [+] Approve (green).	Click the appropriate button based on the decision. For KRISHNAN DEEPAK, click [+] Approve (green). Button click triggers state change and card moves to CO Approved lane.
REG-APP-HEADER	SCR-004-APP-DETAIL	0,0,1440,160	Application header card: Back to Dashboard link (top-right), applicant name (large), app ID, reasoning summary text.	Read applicant name and app ID. Click Back to Dashboard to return to previous screen.
REG-APP-SUMMARY	SCR-004-APP-DETAIL	0,160,1440,340	Application Summary grid with 8 fields in 2 columns: Employer, Income (Declared vs Verified), CIBIL Score, DTI, Risk Category, Decision, Approved Amount, Confidence	Read all 8 fields. For MEHTA VARUN: CIBIL 745, DTI 23.91%, Approved. For PILLAI SAVITA: DTI 73.67%, Rejected. For KRISHNAN DEEPAK: HIL Pending.

Region ID	Applies To	Pixel Bounds x1,y1,x2,y2	Description	Example AI Instruction
			Score.	
REG-TIMELINE	SCR-004-APP-DETAIL	0,340,1440,1000+	Agent Timeline section with TimelineStep nodes (scrollable, extends beyond viewport). Step count is data-driven: 8 steps for clean STP approvals, 9 steps for escalated or rejected cases (includes HIL_ESCALATION). Each step shows: status dot (green/amber/red), tool_name label, step_name heading, optional trigger_reason block (amber), reasoning text, and output_data payload.	Scroll through all steps from top to bottom. Read each step's tool_name, step_name, reasoning, and output_data. Narrate what each step revealed. FINAL_DECISION step output_data contains the terminal outcome — do not assert exact label text, read what is visible.

6. Element Target Registry

The AI should prefer stable selectors and accessible names. Coordinates are fallbacks only — use them when selector-based targeting fails. All coordinates are center (x, y) of the element at 1440 x 900 viewport.

Resilience rules (per Cursor review):

1. Treat applicant names and IDs as preferred targets. If the exact applicant is not found, fall back to any visible card or row in the same lane or table position.

2. Validate by route + heading visibility, not by exact metric values.

3. If a modal or card is not found, refresh the page once, then retry the click.

Element ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Region + Coordinates	Required Action
EL-001	All screens	Overview icon (taskbar)	Link / Icon	a[aria-label="Overview"]	REG-TASKBAR x=676 y=835	Click to navigate to /overview
EL-002	All screens	HIL Review icon (taskbar)	Link / Icon	a[aria-label="HIL Review"]	REG-TASKBAR x=720 y=835	Click to navigate to /hil-review
EL-003	All screens	Applications icon (taskbar)	Link / Icon	a[aria-label="Applications"]	REG-TASKBAR x=764 y=835	Click to navigate to /applications
EL-004	SCR-001-DESKTOP	AI Loan Agent hero title	Text / Read-only	text='AI Loan Agent'	REG-HERO-CENTER x=720 y=360	Read and confirm desktop has loaded. Narrate system introduction.
EL-005	SCR-001-DESKTOP	Queue Snapshot — Need Review tile	Metric tile / Read-only	.grid.grid-cols-2 > div:nth-child(1)	REG-SNAPSHOT-LEFT x=110 y=170	Read the count value. Treat as example — do not assert exact number.
EL-006	SCR-001-DESKTOP	Queue Snapshot — Accepted tile	Metric tile / Read-only	.grid.grid-cols-2 > div:nth-child(3)	REG-SNAPSHOT-LEFT x=250 y=170	Read the count value. Treat as example — do not assert exact number.
EL-007	SCR-002-OVERVIEW	KPI — Applications Today	KPI Card / Read-only	article:nth-of-type(1) in KPI row	REG-KPI-STRIP x=268 y=44	Read value. Treat as example — do not assert exact number.
EL-008	SCR-002-OVERVIEW	KPI — STP Approved %	KPI Card / Read-only	article:nth-of-type(2) in KPI row	REG-KPI-STRIP x=524 y=44	Read value and trend label.
EL-009	SCR-002-OVERVIEW	KPI — HIL Queue	KPI Card / Read-only	article:nth-of-type(3) in KPI row	REG-KPI-STRIP x=780 y=44	Read value. This is the number of cases awaiting human review.

Element ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Region + Coordinates	Required Action
EL-010	SCR-002-OVERVIEW	KPI — Rejected	KPI Card / Read-only	article:nth-of-type(4) in KPI row	REG-KPI-STRIP x=1036 y=44	Read value and rejection rate label.
EL-011	SCR-002-OVERVIEW	KPI — Avg TAT	KPI Card / Read-only	article:nth-of-type(5) in KPI row	REG-KPI-STRIP x=1292 y=44	Read value. Target is under 15 minutes.
EL-012	SCR-002-OVERVIEW	Decision Distribution chart	Chart canvas / Read-only	canvas inside article at REG-CHARTS-ROW1 position 1	REG-CHARTS-ROW1 x=268 y=215	Read legend: STP Approved (green), Rejected (orange), HIL Pending (blue). Explain distribution.
EL-013	SCR-002-OVERVIEW	Risk Distribution chart	Chart canvas / Read-only	canvas inside article at REG-CHARTS-ROW1 position 2	REG-CHARTS-ROW1 x=720 y=215	Read legend: Low (green), High (orange), Medium (red). Explain risk spread.
EL-014	SCR-002-OVERVIEW	Employer Split chart	Chart canvas / Read-only	canvas inside article at REG-CHARTS-ROW1 position 3	REG-CHARTS-ROW1 x=1172 y=215	Read legend: Private (blue), Government (purple). Explain employer mix.
EL-015	SCR-002-OVERVIEW	Daily Batch Volume chart	Bar chart / Read-only	canvas inside article lg:col-span-6 position 1	REG-CHARTS-ROW2 x=358 y=430	Read Mon-Today bars. Explain volume trend and processed vs rejected split.
EL-016	SCR-002-OVERVIEW	Top Rejection Reasons chart	Horizontal bar chart / Read-only	canvas inside article lg:col-span-6 position 2	REG-CHARTS-ROW2 x=1082 y=430	Read bars: HIGH_DTI (longest), LOW_CIBIL, HIGH_RISK, PROXY_SCORE. Explain that DTI is the dominant rejection reason.
EL-017	SCR-002-OVERVIEW	Pipeline strip — all 7 steps	Read-only strip	article with grid-cols-7	REG-PIPELINE x=720 y=590	Read all 7 steps in order: Data Intake, Feature Eng, Rule Engine, LLM Reasoning, HIL Review, Audit Trail, LMS Push. Explain each stage.
EL-018	SCR-002-OVERVIEW	STP Rate metric card	Metric card / Read-only	article in bottom quartet position 1	REG-METRICS-BOTTOM x=198 y=760	Read 57% value and trend note.
EL-019	SCR-002-OVERVIEW	False Reject Rate metric card	Metric card / Read-	article in bottom quartet position 3	REG-METRICS-BOTTOM x=870 y=760	Read 18% value. Explain this means 18% of rejections may be unnecessary — a policy

Element ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Region + Coordinates	Required Action
			only			calibration opportunity.
EL-020	SCR-003-APPLICATIONS	Search input	Input	input[placeholder='Search applicant or app id']	REG-FILTER-BAR x=1050 y=132	Type applicant name or app ID to filter table rows.
EL-021	SCR-003-APPLICATIONS	Status filter dropdown	Select	select:nth-of-type(1) in filter bar	REG-FILTER-BAR x=1188 y=132	Select 'All Status' or specific decision (Conditional Approval, Rejected, HIL Pending).
EL-022	SCR-003-APPLICATIONS	MEHTA VARUN — Open page link	Link	tr:has(text='MEHTA VARUN') a:has-text('Open page') — Playwright-style only; generic fallback: locate row containing text MEHTA VARUN then find anchor with text Open page in same row	REG-APP-TABLE x=1400 y=548	Click to navigate to /applications/APP-2026-00001. Fallback: navigate directly to base_url + /applications/APP-2026-00001.
EL-023	SCR-003-APPLICATIONS	PILLAI SAVITA — Open page link	Link	tr:has(text='PILLAI SAVITA') a:has-text('Open page') — Playwright-style only; generic fallback: locate row containing text PILLAI SAVITA then find anchor with text Open page in same row	REG-APP-TABLE x=1400 y=740	Click to navigate to /applications/APP-2026-00003. Fallback: navigate directly to base_url + /applications/APP-2026-00003.
EL-024	SCR-003-APPLICATIONS	KRISHNAN DEEPAK — Open page link	Link	tr:has(text='KRISHNAN DEEPAK') a:has-text('Open page') — Playwright-style only; generic fallback: locate row containing text KRISHNAN DEEPAK then find anchor with text Open page in same row	REG-APP-TABLE x=1400 y=836	Click to navigate to /applications/APP-2026-00012. Fallback: navigate directly to base_url + /applications/APP-2026-00012.
EL-025	SCR-004-APP-DETAIL	Back to Dashboard link	Link	a:has-text('Back to Dashboard')	REG-APP-HEADER x=1320 y=100	Click to return to /applications table. Use after finishing each application timeline narration.
EL-026	SCR-004-APP-DETAIL	Application Summary grid	Read-only grid	MacWindowCard labeled 'Application Summary'	REG-APP-SUMMARY x=720 y=250	Read all 8 fields: Employer, Income (Declared vs Verified), CIBIL, DTI, Risk, Decision, Approved Amount,

Element ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Region + Coordinates	Required Action
						Confidence.
EL-027	SCR-004-APP-DETAIL	Agent Timeline section	Scrollable list	article:has(h3:has-text('Agent Timeline')) OR div containing heading text 'Agent Timeline' — not standard CSS; use region + heading text match in generic automation	REG-TIMELINE x=720 y=500	Scroll down to reveal all timeline steps. Step count is data-driven per application — expect 8 steps for clean approvals (no HIL_ESCALATION) and 9 steps for escalated or rejected cases (includes HIL_ESCALATION). Read and narrate from KYC_VALIDATION through FINAL_DECISION.
EL-028	SCR-004-APP-DETAIL	Timeline step nodes (each)	Read-only article	article.relative.pl-8 (one per step)	REG-TIMELINE — scroll incrementally 200px per step	For each step: read tool_name (top label, e.g. KYC Service, CIBIL API, System), step_name (heading, e.g. KYC Validation, CIBIL Fetch), status dot color (green=success, amber=warning, red=fail), reasoning text, and output_data payload. If trigger_reason is present (amber block), read it — it explains why this step was triggered. FINAL_DECISION step tool_name is 'System' and output_data contains the terminal outcome. Do not assert exact label text — read what is visible.
EL-029	SCR-005-HIL-REVIEW	HIL gate card header	Read-only card	div or article containing text 'HIL Review - Credit Officer Gate' — use heading text match in region REG-HIL-LANES	REG-HIL-LANES x=720 y=120	Confirm page loaded. Read subtitle: 'Mandatory non-bypassable review queue for flagged applications'.
EL-030	SCR-005-HIL-REVIEW	Awaiting Review lane	Kanban lane / Read-only	div or MacWindowCard with header text 'Awaiting Review'	REG-HIL-LANES x=170 y=300	Read lane header and count badge. Locate KRISHNAN DEEPAK card inside. If card not visible, scroll lane down.
EL-031	SCR-005-HIL-	KRISHNAN DEEPAK kanban	Card / Clicka	div.cursor-pointer:has-	REG-HIL-LANES x=170 y=380	Click to open ApplicationDetailModal.

Element ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Region + Coordinates	Required Action
	REVIEW	card	ble	text('KRISHNAN DEEPAK')		Fallback: click any card in Awaiting Review lane and confirm applicant name in modal header. If modal does not open, refresh page once and retry.
EL-032	SCR-006-HIL-MODAL	Modal title + app ID	Read-only header	header div:has-text('Application Review & Status')	REG-MODAL-HEADER x=480 y=55	Confirm modal opened correctly. Read title and app ID. Verify applicant name matches KRISHNAN DEEPAK.
EL-033	SCR-006-HIL-MODAL	Modal close button	Button	header button:has-text('Close')	REG-MODAL-HEADER x=1180 y=55	Click only if modal needs to be dismissed. Do not click during normal flow.
EL-034	SCR-006-HIL-MODAL	Audit timeline scroll panel	Scrollable section	section.max-h-\[400px\].overflow-y-auto	REG-MODAL-LEFT x=245 y=460	Scroll down to reveal all audit events. Read each event: timestamp, code, step name, description, output snippet. Narrate as you scroll.
EL-035	SCR-006-HIL-MODAL	Risk score display	Read-only	div showing large number score out of 100	REG-MODAL-RIGHT x=790 y=310	Read score value (75 for KRISHNAN DEEPAK). Explain what the score means in the context of the decision.
EL-036	SCR-006-HIL-MODAL	CIBIL Score metric	Metric cell / Read-only	div:has-text('CIBIL Score') in KEY METRICS grid	REG-MODAL-RIGHT x=640 y=540	Read value: 789 (Direct). Explain this is a strong credit score.
EL-037	SCR-006-HIL-MODAL	DTI Ratio metric	Metric cell / Read-only	div:has-text('DTI Ratio') in KEY METRICS grid	REG-MODAL-RIGHT x=950 y=540	Read value: 19.45%. Read badge: 'Within policy'. Explain DTI is well below the 40% limit.
EL-038	SCR-006-HIL-MODAL	Bounce Frequency metric	Metric cell / Read-only	div:has-text('Bounce Frequency') in KEY METRICS grid	REG-MODAL-RIGHT x=640 y=630	Read value: 0/mo. Explain zero bounces means no failed bank transactions — financially clean.
EL-039	SCR-006-HIL-MODAL	Spending Ratio metric	Metric cell / Read-only	div:has-text('Spending Ratio') in KEY METRICS grid	REG-MODAL-RIGHT x=950 y=630	Read value: 0.53. Explain spending ratio of 0.53 means the applicant spends 53% of income — healthy.
EL-040	SCR-006-HIL-MODAL	Cancel button	Button	footer button:has-text('Cancel')	REG-MODAL-FOOTER x=1010 y=843	Click only to dismiss without action. Do not click during normal approval flow.
EL-041	SCR-006-HIL-	Modify & Approve button	Button	footer button.bg-amber-500	REG-MODAL-FOOTER x=1095	Use when approving with modified loan amount.

Element ID	Screen ID	Element Name	Type	Selector / Locator	Fallback Region + Coordinates	Required Action
	MODAL		(amber)		y=843	Not used in this demo flow unless hil_action = modify_approve.
EL-042	SCR-006-HIL-MODAL	Reject button	Button (red)	footer button.bg-rose-600	REG-MODAL-FOOTER x=1195 y=843	Use to reject the application. Not used in this demo flow unless hil_action = reject.
EL-043	SCR-006-HIL-MODAL	Approve button	Button (green)	footer button.bg-emerald-600	REG-MODAL-FOOTER x=1285 y=843	PRIMARY ACTION for this demo. Click after narrating reasoning. After click: confirm modal closes or status updates to CO Approved. Card should move to CO Approved lane in background.

7. Standard AI Action Record Template

Every action the humanoid performs during the demo is recorded in this JSON structure. The blank template below defines all required fields. Five filled examples follow covering the key action types used in this demo flow.

Blank Template:

```
{
  "action_id": "ACT-###",
  "screen_id": "SCR-###",
  "action_type": "click | navigate | read | scroll | think_aloud | explain_screen | validate
| recover",
  "target": {
    "element_id": "EL-###",
    "selector": "preferred stable locator",
    "fallback_region": "REG-...",
    "fallback_coordinates": {"x": 0, "y": 0}
  },
  "input_parameters": {},
  "preconditions": [],
  "execution_steps": [],
  "expected_response": {},
  "validation_rules": [],
  "humanoid_speaker_notes": {
    "before_action": "",
    "during_action": "",
    "after_action": ""
  },
  "failure_recovery": []
}
```

Filled Example 1 — Navigate from Desktop to Overview (action_type: click):

```
{
  "action_id": "ACT-003",
  "screen_id": "SCR-001-DESKTOP",
  "action_type": "click",
  "target": {
    "element_id": "EL-001",
    "selector": "a[aria-label=\"Overview\"]",
    "fallback_region": "REG-TASKBAR",
    "fallback_coordinates": {"x": 676, "y": 835}
  },
  "input_parameters": {},
  "preconditions": [
    "SCR-001-DESKTOP is loaded",
    "Hero card with AI Loan Agent title is visible",
    "Taskbar dock is visible at bottom of viewport"
  ],
  "execution_steps": [
    "Locate Overview icon using selector a[aria-label=\"Overview\"]",
    "Confirm icon is visible and not obscured",
    "Click the icon",
    "Wait for route to change to /overview",
    "Wait for KPI strip to be visible"
  ],
  "expected_response": {
```

```
    "route": "/overview",
    "success_indicator": "KPI strip with 5 cards visible at top of page"
  },
  "validation_rules": [
    "Route must end with /overview",
    "At least one KPI card must be visible — treat values as examples not strict assertions",
    "Three chart cards must be visible below KPI strip"
  ],
  "humanoid_speaker_notes": {
    "before_action": "Let me open the Overview Dashboard to show you today's batch metrics.",
    "during_action": "Navigating to the overview...",
    "after_action": "The Overview Dashboard is now open. I can see the KPI strip and the
charts below."
  },
  "failure_recovery": [
    "If selector not found: use fallback coordinates REG-TASKBAR x=676 y=835",
    "If route does not change within 5s: navigate directly to base_url + /overview",
    "If page still blank: refresh once and retry"
  ]
}
```


Filled Example 2 — Click KRISHNAN DEEPAK Kanban Card (action_type: click):


```
{
  "action_id": "ACT-018",
  "screen_id": "SCR-005-HIL-REVIEW",
  "action_type": "click",
  "target": {
    "element_id": "EL-031",
    "selector": "div.cursor-pointer:has-text(\"KRISHNAN DEEPAK\")",
    "fallback_region": "REG-HIL-LANES",
    "fallback_coordinates": {"x": 170, "y": 380}
  },
  "input_parameters": {
    "applicant_name": "KRISHNAN DEEPAK",
    "application_id": "APP-2026-00012",
    "lane": "Awaiting Review"
  },
  "preconditions": [
    "SCR-005-HIL-REVIEW is loaded",
    "HIL Review - Credit Officer Gate card is visible",
    "Awaiting Review lane is visible with at least one card"
  ],
  "execution_steps": [
    "Locate Awaiting Review lane in REG-HIL-LANES",
    "Find card with text KRISHNAN DEEPAK",
    "If not found: use any card in Awaiting Review lane and confirm name in modal header",
    "Click the card",
    "Wait for ApplicationDetailModal overlay to appear"
  ],
  "expected_response": {
    "modal_visible": true,
    "modal_title": "Application Review & Status",
    "status_label_expected": "HIL PENDING — treat as preferred not strict; assert presence of
any pending/decision badge in modal profile area",
  },
  "validation_rules": [
    "Modal must be visible over the kanban board",
    "Modal header must show Application Review & Status",
    "HIL PENDING status label must be visible on right panel"
  ],
  "humanoid_speaker_notes": {
    "before_action": "I can see KRISHNAN DEEPAK in the Awaiting Review lane. Let me open this
case.",
    "during_action": "Opening the application review modal...",
    "after_action": "The modal is open. I can see the audit trail on the left and the
applicant profile on the right."
```

```
  },
  "failure_recovery": [
    "If card not found by name: click any card in Awaiting Review lane",
    "If modal does not open: refresh /hil-review and retry once",
    "If Awaiting Review lane is empty: narrate and proceed to Applications screen"
  ]
}
```

Filled Example 3 — Scroll and Read Audit Timeline in HIL Modal (action_type: scroll):

```
{
  "action_id": "ACT-019",
  "screen_id": "SCR-006-HIL-MODAL",
  "action_type": "scroll",
  "target": {
    "element_id": "EL-034",
    "selector": "section.max-h-[400px].overflow-y-auto",
    "fallback_region": "REG-MODAL-LEFT",
    "fallback_coordinates": {"x": 245, "y": 460}
  },
  "input_parameters": {
    "scroll_direction": "down",
    "scroll_increment_px": 120,
    "narrate_each_event": true
  },
  "preconditions": [
    "SCR-006-HIL-MODAL is open",
    "Applicant confirmed as KRISHNAN DEEPAK or fallback",
    "Audit timeline section visible in left panel"
  ],
  "execution_steps": [
    "Locate audit timeline using selector section.max-h-[400px].overflow-y-auto",
    "Read first event: timestamp, event_code, title, description, output snippet",
    "Narrate the event",
    "Scroll down 120px",
    "Repeat read and narrate for each event until all events visible",
    "Confirm final event output_data contains terminal decision"
  ],
  "expected_response": {
    "event_codes_seen": ["EVT_DEDUP", "EVT_UNKNOWN", "EVT_FEATURES"],
    "all_events_narrated": true
  },
  "validation_rules": [
    "At least one event with EVT_DEDUP code must be visible",
    "All visible events must be read before scrolling past",
    "Do not skip any event — each tells part of the AI reasoning story",
    "Treat event count as data-driven — do not assert exact number"
  ],
  "humanoid_speaker_notes": {
    "before_action": "Let me scroll through the complete audit trail the AI agent generated
for this applicant.",
    "during_action": "Reading each step — KYC, document analysis, income verification, bank
analysis, CIBIL check...",
    "after_action": "I have reviewed the full audit trail. Income mismatch was flagged but
CIBIL and bank data are both strong."
  },
  "failure_recovery": [
    "If scroll section not found: scroll REG-MODAL-LEFT region directly",
    "If events not visible: wait 2 seconds for API data then retry",
    "If only partial events visible: scroll to bottom and read all visible"
  ]
}
```

Filled Example 4 — Click Approve Button in HIL Modal (action_type: click):

```
{
  "action_id": "ACT-020",
  "screen_id": "SCR-006-HIL-MODAL",
  "action_type": "click",
  "target": {
    "element_id": "EL-043",
    "selector": "footer button.bg-emerald-600",
    "fallback_region": "REG-MODAL-FOOTER",
    "fallback_coordinates": {"x": 1285, "y": 843}
  },
  "input_parameters": {
    "hil_action": "approve",
    "application_id": "APP-2026-00012"
  },
  "preconditions": [
    "SCR-006-HIL-MODAL is open",
    "Full audit trail has been read and narrated",
    "All Key Metrics read: CIBIL 789, DTI 19.45%, Bounce 0/mo, Spending 0.53",
    "Think-aloud reasoning has been delivered",
    "hil_action parameter is confirmed as approve"
  ],
  "execution_steps": [
    "Confirm hil_action parameter = approve before clicking",
    "Locate Approve button using selector footer button.bg-emerald-600",
    "Confirm button color is green and text contains Approve",
    "Deliver final spoken reasoning",
    "Click the Approve button",
    "Wait for modal to close or status to update",
    "Confirm card moved to CO Approved lane"
  ],
  "expected_response": {
    "modal_closes": true,
    "card_lane_after": "CO Approved",
    "application_status_updated": true
  },
  "validation_rules": [
    "Only click if hil_action = approve — check parameter before acting",
    "Button must be green bg-emerald-600 — do not click red or amber buttons",
    "After click: confirm modal closed or status changed",
    "If hil_action = reject: use EL-042 (bg-rose-600) instead",
    "If hil_action = modify_approve: use EL-041 (bg-amber-500) instead"
  ],
  "humanoid_speaker_notes": {
    "before_action": "CIBIL is 789, DTI is 19.45% well within policy, zero bounces, healthy
spending. This is a documentation gap not a credit risk. I am approving this application.",
    "during_action": "Clicking approve...",
    "after_action": "Approved. The application has moved to CO Approved. KRISHNAN DEEPAK will
be notified and the loan moves to disbursement. Decision is logged."
  },
  "failure_recovery": [
    "If button not found by selector: use fallback coordinates x=1285 y=843",
    "If modal does not close after click: wait 3s and check if lane updated",
    "If neither closes nor updates: refresh /hil-review and verify card moved",
    "If action fails completely: narrate intended action and continue flow"
  ]
}
```

Filled Example 5 — Scroll and Narrate Agent Timeline on Application Detail Page (action_type: scroll):

```
{
  "action_id": "ACT-025",
  "screen_id": "SCR-004-APP-DETAIL",
  "action_type": "scroll",
  "target": {
    "element_id": "EL-027",
```

```
    "selector": "div or article containing heading text Agent Timeline — use region REG-
TIMELINE + heading text match; MacWindowCard:has-text is Playwright-only syntax",
    "fallback_region": "REG-TIMELINE",
    "fallback_coordinates": {"x": 720, "y": 500}
  },
  "input_parameters": {
    "scroll_direction": "down",
    "scroll_increment_px": 200,
    "narrate_each_step": true,
    "application_id": "APP-2026-00001"
  },
  "preconditions": [
    "SCR-004-APP-DETAIL is loaded for MEHTA VARUN (APP-2026-00001)",
    "Application Summary has been read",
    "Agent Timeline section is visible"
  ],
  "execution_steps": [
    "Locate Agent Timeline MacWindowCard",
    "Read first TimelineStep: tool_name, step_name, status dot color, reasoning,
output_data",
    "If trigger_reason amber block is present: read it before narrating",
    "Narrate the step",
    "Scroll down 200px",
    "Repeat for each step until FINAL_DECISION step is read and narrated",
    "Deliver closing summary after all steps narrated"
  ],
  "expected_response": {
    "all_steps_narrated": true,
    "final_step_name": "FINAL_DECISION — preferred/expected value; treat as soft check not
hard fail",
    "final_tool_name": "System — preferred/expected; may vary by backend mapping; read what
is visible",
    "terminal_outcome_read": true
  },
  "validation_rules": [
    "Step count is data-driven — do not assert exactly 8 or 9 steps",
    "Every visible step must be narrated before scrolling past it",
    "Read trigger_reason block if present — it explains conditional step execution",
    "FINAL_DECISION output_data contains terminal outcome — read what is visible"
  ],
  "humanoid_speaker_notes": {
    "before_action": "Let me walk you through the complete decision journey the AI agent took
for MEHTA VARUN, step by step.",
    "during_action": "Reading each step — KYC, document analysis, income verification, bank
analysis, CIBIL, feature engineering, decision engine, final decision...",
    "after_action": "Every step came back clean. Income matched perfectly. Strong CIBIL. DTI
well within policy. The Decision Engine approved with 100 percent confidence. No human review
needed."
  },
  "failure_recovery": [
    "If timeline section not visible: scroll page down 300px and retry",
    "If steps not rendered after 3s: refresh and re-navigate to /applications/APP-2026-
00001",
    "If a step is partially cut off: scroll incrementally until full step visible before
narrating"
  ]
}
```

8. Detailed Examples for All Action Types

This table documents every action the humanoid performs during the complete demo flow. Actions are ordered sequentially across all 9 Acts. Treat KPI values and applicant names as preferred targets not strict assertions per Cursor review guidance.

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
ACT-001	Open URL / launch session	Browser navigation	SCR-001-DESKTOP	Navigate to base_url (http://134.33.132.134/maria-aegis-fe)	base_url, session_id	Desktop home loads. Hero card with AI Loan Agent title visible. Taskbar dock visible at bottom.	I will open the Loan Underwriting AI Agent Dashboard.
ACT-002	Read desktop snapshot	Explain screen	SCR-001-DESKTOP	Read REG-SNAPSHOT-LEFT (Queue Snapshot tiles) and REG-DECISION-MIX (Decision Mix bars). Read hero card title.	none	AI Loan Agent title visible. Queue Snapshot tiles visible with counts. Decision Mix bars visible.	This is the Loan Underwriting AI Agent Dashboard. The AI agent automates loan decisions across a 7-step pipeline. Clean cases are auto-approved, policy breaches are auto-rejected, and ambiguous cases are escalated to a human credit officer for review.
ACT-003	Click Overview icon	Mouse click	SCR-001-DESKTOP	Click Overview AppIcon using EL-001 (a[aria-label="Overview"]) in taskbar	element_id: EL-001	Route changes to /overview. KPI strip with 5 cards visible at top.	Let me open the Overview Dashboard to show you today's batch metrics.
ACT-004	Read KPI strip	Data extraction	SCR-002-OVERVIEW	Read all 5 KPI cards EL-007 to EL-011: Applications Today, STP Approved %, HIL Queue, Rejected, Avg TAT	element_ids: EL-007 to EL-011	5 KPI cards visible. Values readable. Treat numbers as live values — read from	Today's batch: [read Applications Today] applications processed. [read STP Approved %]

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
						screen, do not assert exact numbers.	percent were auto-approved by the AI with no human needed. [read HIL Queue] are in the HIL queue awaiting credit officer review. [read Rejected] were rejected. Average turnaround is [read Avg TAT] against a target of under 15 minutes.
ACT-005	Explain charts	Explain screen	SCR-002-OVERVIEW	Read and explain all 5 charts: Decision Distribution (EL-012), Risk Distribution (EL-013), Employer Split (EL-014), Daily Batch Volume (EL-015), Top Rejection Reasons (EL-016)	element_ids: EL-012 to EL-016	All chart canvases visible. Legends readable. HIGH_DTI bar is longest in rejection reasons chart.	The Decision Distribution shows the split between approvals, rejections, and escalations. Risk Distribution shows the risk tier spread. Employer Split shows private vs government borrowers. Daily batch volume shows this is a normal day. Top rejection reason is HIGH_DTI — debt-to-income ratio is the primary killer for most rejected applications.
ACT-006	Explain pipeline strip	Explain screen	SCR-002-OVERVIEW	Read all 7 steps in pipeline strip EL-017: Data Intake, Feature Eng, Rule Engine, LLM Reasoning, HIL Review, Audit Trail, LMS Push	element_id: EL-017	All 7 pipeline step labels visible. Steps are in order 01 through 07.	Every application flows through these 7 stages. Data is ingested, features are engineered,

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
							policy rules are checked, the LLM reasons about edge cases, flagged cases go to HIL Review, every decision is logged in the audit trail, and approved loans are pushed to the LMS for disbursement.
ACT-007	Explain metric cards	Explain screen	SCR-002-OVERVIEW	Read all 4 bottom metric cards: STP Rate (EL-018), Decision Accuracy, False Reject Rate (EL-019), LMS Push Success	element_ids: EL-018, EL-019	All 4 metric cards visible with values and subtitles.	STP Rate of 57 percent means more than half of applications are decided without any human involvement. Decision Accuracy of 85 percent means the model is right 85 percent of the time. False Reject Rate of 18 percent is the cost of being cautious — we are rejecting some applicants who could have been approved. LMS Push Success of 100 percent means zero downstream delivery failures.
ACT-008	Click HIL Review icon	Mouse click	SCR-002-OVERVIEW	Click HIL Review AppIcon using EL-002 (a[aria-label="HIL Review"]) in taskbar	element_id: EL-002	Route changes to /hil-review. HIL Review - Credit Officer Gate card visible.	Now let me show you the cases that need human judgment.
ACT-009	Read all 4 lane	Explain screen	SCR-005-HIL-	Read gate card header EL-029. Read	element_id: EL-029,	Gate card visible with	The HIL Review board has four

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
	headers		REVIEW	all 4 lane headers and count badges in REG-HIL-LANES: Awaiting Review, Under Review, CO Approved, CO Rejected	REG-HIL-LANES	subtitle. 4 lane headers visible with count badges. Treat counts as examples.	lanes. Awaiting Review has one case — nobody has touched it yet. Under Review has two cases being actively reviewed. CO Approved is empty right now. CO Rejected has five cases that credit officers have already declined. Let me open the case that is waiting for me.
ACT-010	Click KRISHNAN DEEPAK kanban card	Mouse click	SCR-005-HIL-REVIEW	Click card in Awaiting Review lane using EL-031 (div.cursor-pointer:has-text("KRISHNAN DEEPAK")). Fallback: any card in Awaiting Review lane.	element_id: EL-031, applicant_name: KRISHNAN DEEPAK	Modal overlay visible. Modal title shows Application Review & Status. Applicant name visible in right panel.	I can see KRISHNAN DEEPAK in the Awaiting Review lane. Let me open this case.
ACT-011	Read modal header and validate	Validate	SCR-006-HIL-MODAL	Read modal title (EL-032), app ID, and applicant name in right panel. Confirm pending/decision badge visible.	element_id: EL-032	Modal title = Application Review & Status. App ID matches application_3_id. Pending/decision badge visible in profile area — treat exact label text as soft check.	The modal is open. This is the Application Review and Status panel for KRISHNAN DEEPAK, application APP-2026-00012. I can see the audit trail on the left and the applicant profile on the right.
ACT-012	Scroll and read full audit timeline	Scroll	SCR-006-HIL-MODAL	Scroll audit timeline section EL-034 (section.max-h-[400px].overflow-y-auto) downward 120px increments. Read and narrate	element_id: EL-034, scroll_increment_px: 120	All audit events visible and narrated. Event codes EVT_DEDUP, EVT_UNKNOWN,	Let me walk through the complete audit trail. KYC passed — identity confirmed.

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
				each event: timestamp, event_code, title, description, output snippet.		EVT_FEATURES seen. Event count is data-driven — do not assert exact number.	Document analysis completed — high confidence. Income verification flagged a mismatch of 27,655 rupees between declared and verified income. Bank statement analysis was triggered by that mismatch — bounce count is zero, spending ratio is healthy. CIBIL score is 789 — genuinely strong credit history. Feature engineering computed DTI at 19.45 percent — well within the 40 percent policy limit. Decision Engine set this to REVIEW at 75 percent confidence — not certain enough to auto-decide.
ACT-013	Read all key metrics	Data extraction	SCR-006-HIL-MODAL	Read all 4 metric cells in KEY METRICS grid: CIBIL Score (EL-036), DTI Ratio (EL-037), Bounce Frequency (EL-038), Spending Ratio (EL-039). Read risk score (EL-035).	element_ids: EL-035 to EL-039	All 4 metric cells readable. Risk score number visible. DTI badge (Within policy) visible. Treat values as examples.	Risk score is 75 out of 100. CIBIL is 789 — excellent credit history. DTI is 19.45 percent and marked within policy. Bounce frequency is zero — no failed transactions. Spending ratio is 0.53 — healthy.

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
ACT-014	Think aloud — deliberate reasoning	Think aloud	SCR-006-HIL-MODAL	Humanoid narrates internal reasoning out loud before deciding. No UI action — pure narration.	hil_action parameter value	No UI validation — this is narration only. hil_action parameter must be confirmed before ACT-015.	The numbers look solid across the board. The income mismatch created uncertainty but the bank data tells a different story — zero bounces, healthy balance, low spending. CIBIL is 789. DTI is 19.45 percent. This looks like a documentation gap, not a credit risk. NTPC is a government employer — salary structures can be complex with allowances. I am going to approve this application.
ACT-015	Click Approve button	Mouse click	SCR-006-HIL-MODAL	Click Approve button EL-043 (footer button.bg-emerald-600). Confirm hil_action = approve before clicking.	element_id: EL-043, hil_action: approve	Modal closes or status updates. Card moves to CO Approved lane. Only click green button — never click red or amber unless hil_action specifies reject or modify_approve.	Approving now.
ACT-016	Respond to user interrupt and narrate	Think aloud	SCR-006-HIL-MODAL	User says: show me some recent applications starting with the approved ones. Humanoid	user_utterance: show me some recent application	No UI validation. Narration confirms approval and	Approved. KRISHNAN DEEPAK has been moved to CO Approved.

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
	transition			narrates approval result then transitions.	s	acknowledges user request.	The loan moves to disbursement and the decision is logged in the audit trail. Of course — let me show you the full application register. I will start with a clean auto-approval so you can see what the happy path looks like.
ACT-017	Click Applications icon	Mouse click	SCR-006-HIL-MODAL	Click Applications AppIcon using EL-003 (a[aria-label="Applications"]) in taskbar	element_id: EL-003	Route changes to /applications. Application register table with rows visible.	Navigating to the applications register.
ACT-018	Read applications table overview	Explain screen	SCR-003-APPLICATIONS	Read filter bar (EL-020, EL-021), read three mini-charts in REG-MINI-CHARTS, read table header row in REG-APP-TABLE.	element_ids: EL-020, EL-021	Table visible with rows (count is data-driven). Filter bar visible. Three mini-charts visible.	Here is the full application register. [Read count from header badge] applications are shown. I can filter by status or risk tier, search by name or app ID. The charts above summarise the same distribution we saw on the overview. Let me open the clean approval first.
ACT-019	Click Open page — MEHTA VARUN	Navigation click	SCR-003-APPLICATIONS	Click Open page link for MEHTA VARUN using EL-022. Preferred: Playwright tr:has-text + a:has-text. Generic fallback: locate row by applicant name text,	element_id: EL-022, application_1_id: APP-2026-00001	Route changes to /applications/APP-2026-00001. Applicant name MEHTA VARUN visible in	Here is MEHTA VARUN — a clean auto-approval. Let me show you the complete decision journey.

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
				then click anchor text Open page in same row. Last resort: navigate directly to base_url + /applications/APP-2026-00001.		header.	
ACT-020	Read application summary — MEHTA VARUN	Data extraction	SCR-004-APP-DETAIL	Read Application Summary grid EL-026: Employer, Income Declared vs Verified, CIBIL, DTI, Risk, Decision, Approved Amount, Confidence.	element_id: EL-026, application_1_id: APP-2026-00001	All 8 summary fields readable. Decision shows Conditional Approval. Confidence shows 100%. Treat values as examples.	MEHTA VARUN works at Reliance Industries. Low risk. Decision is Conditional Approval. Approved amount is 2,29,330 rupees. CIBIL is 745. DTI is 23.91 percent. Confidence is 100 percent — the AI was certain.
ACT-021	Scroll and narrate full timeline — MEHTA VARUN	Scroll	SCR-004-APP-DETAIL	Scroll Agent Timeline EL-027 downward 200px increments. Read and narrate each TimelineStep EL-028: tool_name, step_name, status dot color, reasoning, output_data. Read trigger_reason if present.	element_ids: EL-027, EL-028, scroll_increment_px: 200	All timeline steps narrated. FINAL_DECISION step output_data read. Step count is data-driven — do not assert exact number.	KYC passed. Document analysis high confidence. Income verification — declared income matches verified income perfectly, zero mismatch. Bank analysis shows zero bounces, healthy balance. CIBIL 745. DTI 23.91 percent. Decision Engine approved with 100 percent confidence. No HIL needed. FINAL_DECISION is approved. Every single step came back clean. This is the system working exactly

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
							as designed.
ACT-022	Click Back to Dashboard	Mouse click	SCR-004-APP-DETAIL	Click Back to Dashboard link EL-025 (a:has-text('Back to Dashboard'))	element_id: EL-025	Route returns to /applications. Applications table visible.	Let me go back and show you a pending case now.
ACT-023	Click Open page — KRISHNAN DEEPAK	Navigation click	SCR-003-APPLICATIONS	Click Open page link for KRISHNAN DEEPAK using EL-024. Preferred: Playwright tr:has-text + a:has-text. Generic fallback: locate row by applicant name text, then click anchor text Open page in same row. Last resort: navigate directly to base_url + /applications/APP-2026-00012.	element_id: EL-024, application_3_id: APP-2026-00012	Route changes to /applications/APP-2026-00012. Applicant name KRISHNAN DEEPAK visible in header.	This is KRISHNAN DEEPAK — the case we just reviewed in the HIL queue. Let me show you why it ended up there.
ACT-024	Read summary + scroll timeline — KRISHNAN DEEPAK	Scroll	SCR-004-APP-DETAIL	Read Application Summary EL-026. Then scroll Agent Timeline EL-027 downward 200px increments. Read and narrate each TimelineStep EL-028 including HIL_ESCALATION step.	element_ids: EL-026, EL-027, EL-028, application_3_id: APP-2026-00012	Summary fields readable. Decision shows HIL Pending. All timeline steps narrated including HIL_ESCALATION step. FINAL_DECISION output_data read.	Decision is HIL Pending. The timeline shows KYC passed, documents passed, income mismatch flagged, bank analysis triggered, CIBIL 789, DTI 19.45 percent. All good individually — but the income mismatch created uncertainty. The Decision Engine said I am 75 percent confident, which is below the auto-approval threshold. So the HIL Orchestrator escalated it to a credit officer. That is why it was in my

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
							queue. I reviewed it and approved it. The system worked exactly as intended.
ACT-025	Click Back to Dashboard	Mouse click	SCR-004-APP-DETAIL	Click Back to Dashboard link EL-025 (a:has-text('Back to Dashboard'))	element_id: EL-025	Route returns to /applications. Applications table visible.	Now let me show you a hard rejection.
ACT-026	Click Open page — PILLAI SAVITA	Navigation click	SCR-003-APPLICATIONS	Click Open page link for PILLAI SAVITA using EL-023. Preferred: Playwright tr:has-text + a:has-text. Generic fallback: locate row by applicant name text, then click anchor text Open page in same row. Last resort: navigate directly to base_url + /applications/APP-2026-00003.	element_id: EL-023, application_2_id: APP-2026-00003	Route changes to /applications/APP-2026-00003. Applicant name PILLAI SAVITA visible in header.	This is PILLAI SAVITA — a hard rejection. Notice the header already tells you why: DTI of 73.7 percent exceeds the 40 percent threshold.
ACT-027	Read summary + scroll full timeline — PILLAI SAVITA	Scroll	SCR-004-APP-DETAIL	Read Application Summary EL-026. Then scroll Agent Timeline EL-027 downward 200px increments. Read and narrate each TimelineStep EL-028 including HIL_ESCALATION and FINAL_DECISION.	element_ids: EL-026, EL-027, EL-028, application_2_id: APP-2026-00003	Summary fields readable. Decision shows Rejected. Approved Amount shows Not approved. All timeline steps narrated. FINAL_DECISION output_data read.	PILLAI SAVITA works at BHEL. High risk. Decision is Rejected. Approved amount is not approved. The timeline: KYC passed. Documents passed. Income mismatch of 9,607 rupees flagged. Bank analysis triggered — bounce count 2, spending ratio 0.79, which is high. CIBIL is 763 — actually good. But Feature Engineering computed DTI at 73.67

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
							percent. That is nearly double the 40 percent hard limit. The Decision Engine fired a hard policy breach — automatic rejection. Not a judgment call. Pure mathematics. This person cannot afford this loan at this amount. The system is protecting both the applicant and the lender.
ACT-028	Final closing narration — credit officer authority	Think aloud	SCR-003-APPLICATIONS	After returning from PILLAI SAVITA detail page, humanoid delivers closing narration. No UI action — pure narration. Read live KPI values from SCR-002-OVERVIEW before delivering — do not use hardcoded numbers.	none	No UI validation. Flow complete after narration. Closing metrics must be read from live dashboard not hardcoded.	Throughout this process the AI agent did the legwork — data extraction, risk scoring, policy compliance. But you are the final authority. Every loan leaving this system has been reviewed and approved by a human credit officer. That is mandatory. The AI recommends. You decide. Today: [read Applications Today from KPI strip] applications processed, [read STP Approved count] approved, [read Rejected count] rejected, [read HIL Queue count] awaiting

Action ID	Action Type	Category	Screen	Exact Example	Parameters	Validation	Humanoid Speaker Note
							review. All logged and audited. The system is working as designed. Thank you.

9. Complete Example Flow

This table summarizes all 28 actions in the complete demo sequence, showing action ID, type, screen transition, and expected outcome. Reference this to run the full demo end-to-end.

Action ID	Action Type	From Screen	To Screen	Brief Description	Expected Outcome	Success Indicator
ACT-001	Browser navigation	—	SCR-001-DESKTOP	Open base_url in browser	Desktop home page loads	Hero card visible with AI Loan Agent title
ACT-002	Explain screen	SCR-001-DESKTOP	SCR-001-DESKTOP	Read and narrate desktop snapshot and hero card	User understands system purpose	Agent narrates queue snapshot and decision mix
ACT-003	Click Overview icon	SCR-001-DESKTOP	SCR-002-OVERVIEW	Navigate to overview dashboard	Overview page loads with KPI strip	Route = /overview, KPI cards visible
ACT-004	Read KPI strip	SCR-002-OVERVIEW	SCR-002-OVERVIEW	Read all 5 KPI values from dashboard	Live metric values captured	All 5 KPI cards readable and narrated
ACT-005	Explain charts	SCR-002-OVERVIEW	SCR-002-OVERVIEW	Read and explain 5 charts (Decision, Risk, Employer, Volume, Rejection Reasons)	User sees data distributions	All charts visible and narrated
ACT-006	Explain pipeline	SCR-002-OVERVIEW	SCR-002-OVERVIEW	Read 7-step pipeline strip and narrate each stage	User understands AI workflow	Pipeline strip visible, all 7 steps read
ACT-007	Explain metrics	SCR-002-OVERVIEW	SCR-002-OVERVIEW	Read 4 bottom metric cards (STP Rate, Accuracy, False Reject, LMS Push)	User sees system performance KPIs	All 4 metric cards narrated
ACT-008	Click HIL Review icon	SCR-002-OVERVIEW	SCR-005-HIL-REVIEW	Navigate to HIL review kanban board	Kanban page loads with 4 lanes	Route = /hil-review, gate card visible
ACT-009	Explain lanes	SCR-005-HIL-REVIEW	SCR-005-HIL-REVIEW	Read all 4 lane headers and counts	User sees queue structure	All 4 lanes visible with count badges
ACT-010	Click kanban card	SCR-005-HIL-REVIEW	SCR-006-HIL-MODAL	Click KRISHNAN DEEPAK card in Awaiting Review lane	HIL modal opens	Modal overlay visible, app ID and name confirmed
ACT-011	Validate modal header	SCR-006-HIL-MODAL	SCR-006-HIL-MODAL	Read modal title and applicant profile in right panel	Modal structure confirmed	Application Review & Status title visible

Action ID	Action Type	From Screen	To Screen	Brief Description	Expected Outcome	Success Indicator
ACT-012	Scroll audit trail	SCR-006-HIL-MODAL	SCR-006-HIL-MODAL	Scroll down in left panel, read all audit events	Full timeline narrated	All events visible and narrated
ACT-013	Read key metrics	SCR-006-HIL-MODAL	SCR-006-HIL-MODAL	Read CIBIL, DTI, Bounce, Spending metrics from right panel	Metrics captured for reasoning	All 4 metrics readable
ACT-014	Think aloud	SCR-006-HIL-MODAL	SCR-006-HIL-MODAL	Narrate internal reasoning about approval decision	Humanoid explains decision logic	Narration delivered, no UI change
ACT-015	Click Approve	SCR-006-HIL-MODAL	SCR-006-HIL-MODAL	Click green Approve button	Card moves to CO Approved lane	Modal closes or card lane updates
ACT-016	Respond to interrupt	SCR-006-HIL-MODAL	SCR-006-HIL-MODAL	Narrate approval result and acknowledge user request to show applications	Transition narration delivered	Narration confirms approval + next step
ACT-017	Click Applications icon	SCR-006-HIL-MODAL	SCR-003-APPLICATIONS	Navigate to applications table	Applications page loads	Route = /applications, table visible with 14 rows
ACT-018	Explain table	SCR-003-APPLICATIONS	SCR-003-APPLICATIONS	Read filter bar, mini-charts, and table overview	User sees application register structure	Table visible with rows and filters
ACT-019	Click Open — MEHTA VARUN	SCR-003-APPLICATIONS	SCR-004-APP-DETAIL	Navigate to MEHTA VARUN detail page (APP-2026-00001)	Detail page loads for happy path case	Route = /applications/APP-2026-00001
ACT-020	Read summary — MEHTA VARUN	SCR-004-APP-DETAIL	SCR-004-APP-DETAIL	Read Application Summary grid (8 fields)	Happy path metrics captured	All summary fields readable
ACT-021	Scroll timeline — MEHTA VARUN	SCR-004-APP-DETAIL	SCR-004-APP-DETAIL	Scroll down and narrate all timeline steps to FINAL_DECISION	Happy path reasoning narrated	All steps read, FINAL_DECISION = APPROVED
ACT-022	Click Back	SCR-004-APP-DETAIL	SCR-003-APPLICATIONS	Return to applications table	Table reloaded	Route = /applications, table visible
ACT-023	Click Open — KRISHNAN DEEPAK	SCR-003-APPLICATIONS	SCR-004-APP-DETAIL	Navigate to KRISHNAN DEEPAK detail page (APP-2026-00012)	Detail page loads for pending case	Route = /applications/APP-2026-00012
ACT-024	Scroll timeline — KRISHNAN DEEPAK	SCR-004-APP-DETAIL	SCR-004-APP-DETAIL	Read summary + scroll and narrate all timeline steps including	Pending case reasoning narrated	All steps read including HIL_ESCALATION,

Action ID	Action Type	From Screen	To Screen	Brief Description	Expected Outcome	Success Indicator
				HIL_ESCALATION		FINAL_DECISION = REVIEW
ACT-025	Click Back	SCR-004-APP-DETAIL	SCR-003-APPLICATIONS	Return to applications table	Table reloaded	Route = /applications, table visible
ACT-026	Click Open — PILLAI SAVITA	SCR-003-APPLICATIONS	SCR-004-APP-DETAIL	Navigate to PILLAI SAVITA detail page (APP-2026-00003)	Detail page loads for rejection case	Route = /applications/APP-2026-00003
ACT-027	Scroll timeline — PILLAI SAVITA	SCR-004-APP-DETAIL	SCR-004-APP-DETAIL	Read summary + scroll and narrate all timeline steps to FINAL_DECISION	Rejection case reasoning narrated	All steps read, FINAL_DECISION = REJECTED, DTI breach confirmed
ACT-028	Final narration	SCR-004-APP-DETAIL	SCR-003-APPLICATIONS	Return and deliver closing remarks about credit officer authority	Demo concludes with key message	Narration emphasizes human final authority + system logging

10. Detailed Page Example: Loan Agent Desktop (SCR-001-DESKTOP)

The desktop landing page serves as the entry point for the demo. This section maps every visible UI region and explains how the humanoid reads and interprets the desktop to begin the walkthrough.

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
Background	0,0,1440,900	Desktop environment aesthetic — light blue gradient background	Confirm desktop has loaded by seeing the background clear and visible. No error messages.	Background rendered without errors
Top-left: Queue Snapshot panel	20,30,380,280	Shows four metric tiles: Need Review (amber), Reviewed (blue), Accepted (green), Rejected (red). Each tile displays count and label.	Read all four tiles. Today's queue snapshot tells me the distribution of applications. Need Review is how many are awaiting my attention. Reviewed is work already done. Accepted and Rejected are final decisions.	All 4 tiles visible and readable. Treat counts as live values — do not assert exact numbers.
Queue Snapshot — Need Review tile	45,60,180,130	Amber tile showing applications requiring human review	Read the count in this tile. This is the HIL queue size.	Count visible and readable
Queue Snapshot — Reviewed tile	190,60,325,130	Blue tile showing applications already processed by a credit officer	Read the count. This shows how much work the team	Count visible and readable

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
			completed today.	
Queue Snapshot — Accepted tile	45,140,180,210	Green tile showing approved applications	Read the count. This is the approval rate at a glance.	Count visible and readable
Queue Snapshot — Rejected tile	190,140,325,210	Red tile showing rejected applications	Read the count. This is the rejection rate at a glance.	Count visible and readable
Top-right: Decision Mix panel	1060,30,1410,280	Shows three progress bars: Accepted (green), Rejected (red), Under Review (orange). Each bar shows count and percentage.	Read the decision mix. Accepted bar shows approval ratio. Rejected bar shows rejection ratio. Under Review bar shows cases still pending decision.	All 3 bars visible with percentages readable
Decision Mix — Accepted bar	1220,92,1390,130	Green bar showing accepted application ratio and count	Read the count and percentage. This is how many cases are approved.	Percentage and count visible and readable
Decision Mix — Rejected bar	1220,137,1390,170	Red bar showing rejected application ratio and count	Read the count and percentage. This is how many cases are rejected.	Percentage and count visible and readable
Decision Mix — Under Review bar	1220,181,1390,215	Orange bar showing cases awaiting human review	Read the count and percentage. This is how many are in the queue.	Percentage and count visible and readable
Center: Hero card	310,260,1130,520	Large Mac-style window with	Confirm the hero card is	Hero card rendered with correct styling and text

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
		red/yellow/green chrome dots at top. Title: LOAN UNDERWRITING AI AGENT. Heading: AI Loan Agent. Subtitle: Virtual Credit Underwriter.	visible. This is the main system title and confirms the dashboard has loaded. The macOS chrome styling is intentional — this is a demo of how the AI operates a real interface.	
Hero card — Chrome dots	348,290,366,302	macOS chrome buttons (red close, yellow minimize, green maximize)	Notice the macOS styling. This signals that the interface is a simulation of a real application window.	Chrome dots visible at top-left of card
Hero card — Main title	667,329,775,339	Small label: LOAN UNDERWRITING AI AGENT	Read this label — it confirms the system name.	Label text visible
Hero card — Heading	603,379,837,421	Large heading: AI Loan Agent	Read the main heading. This is the system name.	Heading text visible and large
Hero card — Subtitle	673,434,767,444	Subtitle: Virtual Credit Underwriter	Read the subtitle. This explains the AI's role.	Subtitle text visible and readable
Bottom taskbar	520,620,940,700	Three icon buttons (Overview, HIL Review, Applications) plus live clock. Rounded dock-style container.	The taskbar at the bottom shows three navigation options: Overview Dashboard, HIL Review	Taskbar visible, 3 icons visible, clock visible

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
			queue, and Applications register. I will use these icons to navigate throughout the demo.	
Taskbar — Overview icon	600,630,650,680	Blue bar chart icon labeled Overview	This icon opens the Overview Dashboard where I can see KPIs and charts.	Icon visible and clickable
Taskbar — HIL Review icon	660,630,710,680	Orange checklist icon labeled HIL Review	This icon opens the HIL Review kanban board where pending cases are queued.	Icon visible and clickable
Taskbar — Applications icon	720,630,770,680	Green spreadsheet icon labeled Applications	This icon opens the applications table showing all 14 cases in the batch.	Icon visible and clickable
Taskbar — Clock	830,650,890,670	Live time display (HH:MM:SS format)	The clock shows the current time. I will narrate time as part of the system status if needed.	Time visible and updates

11. Detailed Page Example: Overview Dashboard (SCR-002-OVERVIEW)

Route: /overview. Shows KPI strip, five charts, 7-step pipeline, and performance metrics. Agent reads and explains each section.

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
KPI Strip	0,0,1440,90	Five KPI cards: Applications Today, STP Approved %, HIL Queue, Rejected, Avg TAT	Read each card. [Read Applications Today count], [Read STP Approved %], [Read HIL Queue count], [Read Rejected count], [Read Avg TAT]. Treat as live values.	All 5 KPI cards visible and readable
Decision Distribution chart	10,90,470,340	Doughnut chart showing Approved, Rejected, HIL Pending segments with legend	Explain the decision split. Read legend: STP Approved (green segment), Rejected (orange), HIL Pending (blue). Explain distribution.	Chart visible with legend
Risk Distribution chart	480,90,940,340	Doughnut chart showing Low, High, Medium risk tiers with legend	Explain risk spread. Read legend and segments. Low risk (green), High risk (orange), Medium risk (red).	Chart visible with legend
Employer Split chart	950,90,1410,340	Doughnut chart showing Private vs	Explain employer	Chart visible with legend

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
		Government employers with legend	mix. Read segments and percentages .	
Daily Batch Volume chart	10,340,700,540	Bar chart showing Mon-Today with Processed and Rejected bars	Explain volume trend. Compare processed vs rejected bars daily. Identify peaks.	Chart visible with bars
Top Rejection Reasons chart	710,340,1410,540	Horizontal bar chart showing HIGH_DTI, LOW_CIBIL, HIGH_RISK, PROXY_SCORE	Explain rejection reasons. HIGH_DTI should be the longest bar. Read all bars left to right.	Chart visible with bars, HIGH_DTI dominant
Pipeline strip	0,540,1440,640	Seven numbered steps in horizontal row: Data Intake, Feature Eng, Rule Engine, LLM Reasoning, HIL Review, Audit Trail, LMS Push	Read all 7 steps in order. Explain what each stage does in the AI pipeline. This is the complete workflow.	All 7 steps visible and labeled
STP Rate card	0,640,360,820	Metric card showing STP Rate percentage and trend	Read STP Rate value. Explain this is the auto-approval rate without human intervention .	Value visible and readable
Decision Accuracy card	360,640,720,820	Metric card showing Decision Accuracy percentage	Read accuracy value. Explain model correctness rate.	Value visible and readable
False Reject Rate card	720,640,1080,820	Metric card showing False Reject Rate	Read false reject value.	Value visible and readable

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
		percentage	Explain this is unnecessary rejections — opportunity for policy tuning.	
LMS Push Success card	1080,640,1440,820	Metric card showing LMS Push Success percentage	Read LMS success value. Explain this is downstream delivery success rate.	Value visible and readable

12. Detailed Page Example: HIL Review Kanban (SCR-005-HIL-REVIEW)

Route: /hil-review. Shows gate card header and four kanban lanes: Awaiting Review, Under Review, CO Approved, CO Rejected. Agent reads lanes and opens cards.

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
Gate card header	30,10,1200,150	MacWindowCard with title 'HIL Review - Credit Officer Gate' and subtitle about mandatory review	Confirm page loaded. Read gate title. Understand this is a mandatory non-bypassable review queue.	Gate card visible with title
Awaiting Review lane	20,160,360,850	First kanban lane with header 'Awaiting Review' and count badge. Contains cards not yet opened.	Read lane header and count. Look for KRISHNAN DEEPAK card. This is the queue of cases that need immediate attention.	Lane header visible with count badge
Under Review lane	370,160,710,850	Second kanban lane with header 'Under Review' and count badge. Contains cards currently being reviewed.	Read lane header and count. These are cases being actively worked on by credit officers.	Lane header visible with count badge
CO Approved lane	720,160,1060,850	Third kanban lane with header 'CO Approved' and count badge. Contains approved cards.	Read lane header and count. These are cases that credit officers have approved and are	Lane header visible with count badge

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
			ready for disbursement.	
CO Rejected lane	1070,160,1410,850	Fourth kanban lane with header 'CO Rejected' and count badge. Contains rejected cards.	Read lane header and count. These are final rejections by credit officers. No further action.	Lane header visible with count badge
Kanban card (example: KRISHNAN DEEPAK)	varies,160-850	Card showing: app ID, applicant name, loan amount, tenure, risk chip (colored), risk score bar	Click card to open modal. Card displays key info: APP-2026-00012, KRISHNAN DEEPAK, loan amount, risk level, score bar.	Card visible and clickable. Correct applicant name visible

13. Detailed Page Example: HIL Modal (SCR-006-HIL-MODAL)

Modal overlay on /hil-review. Shows Application Review & Status with left audit trail and right applicant profile. Agent scrolls, reads, reasons, and decides.

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
Modal header	30,10,1200,100	Title 'Application Review & Status', app ID, Close button (rose/pink)	Read modal title. Confirm applicant name and app ID. Use Close button only to dismiss without action.	Modal title visible, app ID visible
Left panel: Audit timeline scroll section	30,120,460,800	Scrollable section with audit event cards. Each event shows timestamp, event_code, step name, description, output snippet	Scroll down through all events. Read each event: KYC, Document Analysis, Income Verification, Bank Analysis, CIBIL Fetch, Feature Engineering, Decision Engine, HIL Escalation. Narrate each before scrolling.	Events visible and scrollable. All events narrated
Right panel: Applicant profile	480,120,1200,400	Shows applicant name, app ID, status label (e.g., HIL PENDING), risk score (0-100) with progress bar	Read applicant name and confirm match. Read status badge. Read risk score out of 100.	Name, ID, status, score all visible
Key Metrics grid (4 cells)	480,410,1200,800	Grid showing: CIBIL Score, DTI Ratio, Bounce Frequency,	Read all 4 metrics: [Read CIBIL	All 4 metric cells visible and readable

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
		Spending Ratio. Each cell has label and value.	Score], [Read DTI Ratio + badge], [Read Bounce Frequency], [Read Spending Ratio]. These drive the decision.	
Modal footer	30,820,1200,880	Four action buttons: Cancel (gray/white), Modify & Approve (amber), Reject (red), Approve (green)	Four decision buttons. Approve is green (bg-emerald-600), Reject is red (bg-rose-600), Modify & Approve is amber (bg-amber-500), Cancel is white. Only click the button matching hil_action parameter.	All 4 buttons visible. Colors correct

14. Detailed Page Example: Applications Table (SCR-003-APPLICATIONS)

Route: /applications. Shows filter bar, mini-charts, and application register table with 14 rows. Agent searches for applicants, clicks table rows to trigger ApplicationReviewModal (read-only), and uses 'Open page' links to navigate to detail view.

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
Search input	600,100,1200,165	Text input: 'Search applicant or app id' placeholder. Type to filter table by name or app ID.	Use search to locate applicants by name (MEHTA VARUN, PILLAI SAVITA, KRISHNAN DEEPAK) or app ID (APP-2026-00001, etc.). Search narrows table to matching rows.	Search input visible and functional
Status filter dropdown	1070,100,1200,165	Dropdown: 'All Status' or filter by decision status (Conditional Approval, Rejected, HIL Pending)	Optional: use to filter by decision type. For this demo, keep as 'All Status'.	Dropdown visible
Risk filter dropdown	1210,100,1330,165	Dropdown: 'All Risk Tiers' or filter by risk level (Low, Medium, High)	Optional: use to filter by risk. For this demo, keep as 'All Risk Tiers'.	Dropdown visible
Export CSV button	1340,100,1420,165	Button to export table as CSV	Not used in this demo.	Button visible
Mini-charts row	0,220,1440,380	Three small charts: Decision Distribution, Risk Distribution, Confidence Distribution. Summarises the current batch of applications.	Read the mini-charts. Confirm they match distributions seen in the Overview dashboard.	All 3 mini-charts visible

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
Table header row	0,380,1440,420	Columns: Applicant, Loan Amount, Decision, Risk, Confidence, HIL, Created, Route (Open page link)	Read header to understand table structure. Each row is one application.	Header visible with all columns
Table body — 14 rows	0,420,1440,1000+	Rows showing: MEHTA VARUN (APP-2026-00001, Conditional Approval, Low Risk, 100% confidence), PILLAI SAVITA (APP-2026-00003, Rejected, High Risk), KRISHNAN DEEPAK (APP-2026-00012, HIL Pending, Medium Risk), and 11 others	Locate each applicant by name or app ID. Each row shows decision, risk, confidence. Click row to trigger ApplicationReviewModal (read-only).	All 14 rows visible. Three target applicants visible
Table row click (KRISHNAN DEEPAK)	0,650,1440,680	Clicking any row in the table body	Click row to open ApplicationReviewModal. This is a read-only modal showing audit trail and profile. From here, click the 'Open page' link in the modal to navigate to full detail page.	ApplicationReviewModal appears as overlay
ApplicationReviewModal overlay	50,100,1390,800	Read-only modal showing: Applicant name, App ID, Audit events (left panel), Profile and metrics (right panel). NO action buttons. Contains 'Open page' link.	Read the data in the modal. This is a preview of the full application. Click 'Open page' link to navigate to detail page (/applications/:id).	Modal visible with correct applicant name and app ID
Open page link in modal	1200,750,1340,780	Hyperlink inside ApplicationReviewMo	Click this link to	Link visible in modal footer

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
		dal that navigates to /applications/:id detail page	navigate to the full Application Detail page. Do not close the modal without clicking this link during the demo.	
Open page link — MEHTA VARUN	1400,548,1440,575	Link in Route column for MEHTA VARUN row. Navigates to /applications/APP-2026-00001	Alternative direct path: click 'Open page' in table row instead of going through modal. Both paths lead to same detail page.	Link visible and clickable
Open page link — KRISHNAN DEEPAK	1400,660,1440,687	Link in Route column for KRISHNAN DEEPAK row. Navigates to /applications/APP-2026-00012	Alternative direct path: click 'Open page' in table row instead of going through modal.	Link visible and clickable
Open page link — PILLAI SAVITA	1400,740,1440,767	Link in Route column for PILLAI SAVITA row. Navigates to /applications/APP-2026-00003	Alternative direct path: click 'Open page' in table row instead of going through modal.	Link visible and clickable

15. Detailed Page Example: Application Detail (SCR-004-APP-DETAIL)

Route: /applications/:id. Shows application summary grid and full agent timeline. Agent reads summary, then scrolls and narrates every timeline step to FINAL_DECISION.

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
Back to Dashboard link	1200,80,1400,120	Link in header. Navigates back to /applications table.	Click after finishing timeline narration to return to application list.	Link visible and clickable
Applicant name header	30,80,800,160	Large heading showing applicant name (MEHTA VARUN, PILLAI SAVITA, or KRISHNAN DEEPAK)	Read and confirm applicant name at top of page.	Applicant name visible and large
Application ID header	30,160,400,200	Shows app ID (APP-2026-00001, APP-2026-00003, or APP-2026-00012)	Read app ID to confirm correct application.	App ID visible
Application Summary grid	30,200,1200,340	8-field grid in 2 columns: Employer, Income (Declared vs Verified), CIBIL Score, DTI Ratio, Risk Category, Decision, Approved Amount, Confidence Score	Read all 8 fields. For MEHTA VARUN: Reliance Industries, income match, CIBIL 745, DTI 23.91%, Low risk, Conditional Approval, amount, 100% confidence. For PILLAI SAVITA: BHEL, income mismatch, CIBIL 763, DTI 73.67%, High risk, Rejected. For	All 8 fields readable

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
			KRISHNAN DEEPAK: confirm all values.	
Agent Timeline section header	30,350,400,390	Section titled 'Agent Timeline' with subtitle 'Step-by-step execution trace showing how decision was produced'	Confirm timeline section is visible. This is where the full decision journey is shown.	Timeline section header visible
Timeline steps (scrollable)	30,400,1200,1000+	Series of TimelineStep nodes (data-driven: 8 steps for clean approvals, 9 steps for escalated/rejected cases). Each step shows: status dot (green/amber/red), tool_name (top label), step_name (heading), trigger_reason (amber block if present), reasoning (description), output_data (code block)	Scroll down through all steps. For each step: read tool_name, step_name, reasoning, output_data. If trigger_reason block present (amber), read it. Narrate each step before scrolling to next. Continue until FINAL_DECISION step is visible and read.	All steps visible and scrollable. FINAL_DECISION step at bottom
Timeline step 1 — KYC Validation	30,410,1200,480	Status dot (green=success), step name, KYC Service tool, reasoning, output	Read: KYC passed. Identity confirmed.	Step visible
Timeline step 2 — Document Analysis	30,490,1200,560	Status dot, step name, step reasoning, output	Read: Documents analyzed with high confidence.	Step visible
Timeline step 3 — Income Verification	30,570,1200,640	Status dot (amber=warning if mismatch), step name, step reasoning	Read: Income verified. For MEHTA	Step visible

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
			VARUN: perfect match. For others: flag mismatch amount.	
Timeline step 4 — Bank Analysis	30,650,1200,720	Status dot, step name, Bank Analyzer tool, reasoning with bounce count and spending ratio	Read: Bank balance, bounce count, spending ratio. Assess financial health.	Step visible
Timeline step 5 — CIBIL Fetch	30,730,1200,800	Status dot (green), step name, CIBIL API tool, reasoning, CIBIL score output	Read: CIBIL score and any flags (defaults, etc.).	Step visible
Timeline step 6 — Feature Engineering	30,810,1200,880	Status dot, step name, Policy Rules Engine tool, reasoning with DTI calculation	Read: DTI computed. Check if within policy (40% limit) or breached.	Step visible
Timeline step 7 — Decision Engine	30,890,1200,960	Status dot, step name, Policy Rules Engine tool, reasoning, confidence and decision output	Read: Confidence level (100% = auto-approve, 75% = review needed). Decision: APPROVED, REVIEW, or REJECTED.	Step visible
Timeline step 8 (if present) — HIL Escalation	30,970,1200,1040	Status dot (amber), step name, HIL Orchestrator tool, reasoning, escalation reason	Present only for pending/escalated cases (KRISHNAN DEEPAK, PILLAI SAVITA). Read why escalated to human.	Step visible if applicable
Timeline step 9 — FINAL_DECISION	30,1050,1200,1120	Status dot (final color), step name, System tool,	Final step. Read output_data	Step visible at bottom

UI Area	Location (x1,y1,x2,y2)	Purpose	Humanoid Instruction	Validation
		reasoning, output_data with terminal outcome	. This shows the terminal outcome: APPROVED, REJECTED, or REVIEW. Do not assert exact label text — read what is visible.	

16. Pop-ups and Modal Examples

This section documents modal overlays and pop-up behaviors that the agent may encounter during the demo. All modals are triggered by user interactions (clicks on table rows or kanban cards) and require specific handling.

Modal Type	Trigger	What Agent Must Read	Correct Action	Example Narration	Validation
ApplicationReviewModal (Read-Only)	Clicking any row in /applications table	Modal title, applicant name, app ID, audit trail (left panel), profile and metrics (right panel)	Read all visible data. Click 'Open page' link to navigate to /applications/:id detail page. Do NOT click row again or close without reading.	The application review modal has opened. I can see KRISHNAN DEEPAK's record. The app ID is APP-2026-00012. Let me read the audit trail on the left side... I see KYC passed, documents analyzed, income verification completed... Now I'll click Open page to see the full detail view.	Modal overlay visible, correct applicant name and app ID displayed, 'Open page' link visible
ApplicationDetailModal (HIL Action Modal)	Clicking kanban card in /hil-review 'Awaiting Review' lane	Modal title 'Application Review & Status', applicant name, app ID, audit trail (left), applicant profile (right), key metrics (CIBIL, DTI, Bounce, Spending), three action buttons (Approve/Reject/Modify), Close button	Read all sections. Scroll audit trail if needed. Read all 4 metrics carefully. Think aloud about the decision. Click [Approve] or [Reject] or [Modify & Approve] button only. Do NOT click Close during normal flow.	The HIL modal has opened for KRISHNAN DEEPAK. APP-2026-00012. I can see the audit trail... Reading metrics: CIBIL 789, DTI 19.45%, Bounce 0, Spending 0.53. The decision engine flagged this for review because of the income mismatch. But with 75% confidence and medium risk, I believe this should be approved. Clicking Approve button.	Modal title 'Application Review & Status' visible, all 4 metrics readable, three decision buttons (green/red/amber) visible, modal closes after button click or status updates

Handling Notes for Agent:

1. ApplicationReviewModal is triggered only by clicking table rows on /applications. Direct clicks on 'Open page' links bypass this modal.

2. ApplicationDetailModal (HIL Modal) is triggered ONLY by clicking kanban cards in /hil-review. It is NOT the same as ApplicationReviewModal.

3. If a modal does not appear after a click, refresh the page and retry once. If it still does not appear, use the fallback region coordinates from Section 6 (Element Target Registry).

4. Modal action buttons in ApplicationDetailModal have exact selectors: Approve (bg-emerald-600), Reject (bg-rose-600), Modify & Approve (bg-amber-500), Cancel (white). Match colors before clicking.

5. After clicking any decision button in ApplicationDetailModal, the modal closes and the kanban card should move to the appropriate lane (CO Approved, CO Rejected, etc.). Wait up to 3 seconds for the UI to update before concluding the action failed.

17. Decision Logic and Branching Examples

This section documents all key decision points and branching scenarios the agent may encounter during the demo. Each entry shows the condition, the AI decision, the action mapped to it, and the narration to deliver.

Decision ID	Condition	AI Decision	Action	Narration
DEC-001	Desktop loaded. Queue snapshot and decision mix visible. Next step is overview.	Navigate to overview for full KPI and chart review.	ACT-003: Click Overview icon (blue bar chart) in taskbar	The desktop is loaded. I can see today's queue snapshot on the left and the decision mix on the right. Let me now navigate to the overview dashboard to walk through the full system metrics.
DEC-002	Overview loaded. All KPIs, 5 charts, pipeline strip, and 4 metric cards reviewed.	Navigate to HIL review kanban — cases in queue need attention.	ACT-008: Click HIL Review icon (orange clipboard) in taskbar	I've reviewed the overview. The STP rate and decision distributions look healthy. There are cases waiting in the HIL queue for my decision. Let me go to the HIL review board.
DEC-003	HIL kanban loaded. KRISHNAN DEEPAK card visible in Awaiting Review lane.	Click card to open HIL modal and review application before deciding.	ACT-010: Click KRISHNAN DEEPAK kanban card	I can see the kanban board with four lanes. KRISHNAN DEEPAK's application is in the Awaiting Review lane. Let me open it to review the details.
DEC-004	HIL modal open. Audit trail scrolled and narrated. All 4 metrics read: CIBIL 789, DTI 19.45%, Bounce 0, Spending 0.53. Confidence 75%, Medium risk, income mismatch flagged.	DTI within policy limit. CIBIL strong. Mismatch within acceptable range. Approve.	ACT-015: Click Approve button (bg-emerald-600)	I've reviewed the full audit trail and all four key metrics. The CIBIL score of 789 is strong. DTI of 19.45% is well within the 40% policy limit. Bounce is zero and spending ratio is moderate. The income mismatch is flagged but within tolerance. I'm satisfied that this application should be approved.
DEC-005	Approve button clicked. Modal closes. KRISHNAN DEEPAK card moves to CO Approved lane. User asks to see the full applications register.	Navigate to applications table to show full batch.	ACT-017: Click Applications icon (green spreadsheet) in taskbar	The approval has been recorded. KRISHNAN DEEPAK's card has moved to the CO Approved lane. Now let me navigate to the applications table to show the complete batch

Decision ID	Condition	AI Decision	Action	Narration
				register.
DEC-006	Applications table loaded. Target applicant row visible.	Click table row to open ApplicationReviewModal for a preview before opening full detail.	ACT-019/023/026: Click target row in table	Here is the full applications register. Let me click on this row to open the application preview.
DEC-007	ApplicationReviewModal open (read-only). Audit trail and profile visible. No action buttons.	Read available data. Then navigate to full detail page via Open page link.	Click Open page link in modal	The review modal is showing me the application summary. This is a read-only preview. I can see the audit trail and key profile data. Let me click Open page to go to the full detail view.
DEC-008	Application Detail page loaded. Summary grid visible. Timeline section visible.	Read all 8 summary fields first, then scroll timeline step by step.	ACT-020/024/027: Read summary then scroll timeline	I'm on the application detail page. Let me first read the summary grid — employer, declared and verified income, CIBIL score, DTI ratio, risk category, decision, approved amount, confidence score. Now let me scroll through the timeline.
DEC-009	Scrolling timeline. Step count is 9 (HIL_ESCALATION present — escalated or rejected case).	This is an escalated case. Read HIL_ESCALATION step and explain why it triggered.	Scroll to HIL_ESCALATION step and narrate trigger_reason block	I can see this application has 9 timeline steps including a HIL Escalation step. Let me read the trigger reason — this explains why the system escalated this case for human review instead of auto-deciding.
DEC-010	Scrolling timeline. Step count is 8 (no HIL_ESCALATION — clean STP case).	This is a straight-through auto-approved case. No escalation step.	Scroll directly to FINAL_DECISION after Decision Engine step	This application has 8 timeline steps with no HIL escalation. The Decision Engine had sufficient confidence to auto-decide without human review.
DEC-011	FINAL_DECISION step visible at bottom of timeline.	Read output_data. Do not assert exact label — read what is visible on screen.	ACT-021/024/027: Read FINAL_DECISION output_data	I've reached the final decision step. Let me read the outcome as shown on screen. [Read visible output_data text]. This is the terminal outcome of the agent's full pipeline.
DEC-012	Timeline fully narrated. Demo requires returning to applications table.	Click Back to Dashboard link to return to /applications.	Click Back to Dashboard link in header (x=1200-1400, y=80-120)	I've completed the full timeline walkthrough for this application. Let me click Back to Dashboard

Decision ID	Condition	AI Decision	Action	Narration
				to return to the applications register.
DEC-013	Kanban card for KRISHNAN DEEPAK not found in Awaiting Review lane.	Refresh the HIL review page once and retry. If still missing, check Under Review lane.	Refresh /hil-review, then look in Under Review lane	I don't see KRISHNAN DEEPAK's card in the Awaiting Review lane. Let me refresh the page and check again. If it has moved to Under Review, I'll open it from there.
DEC-014	HIL modal does not open after clicking kanban card.	Retry click once. If modal still does not appear, use fallback selector from Element Registry.	Retry click on EL-027 (kanban card selector)	The modal did not open on the first click. Let me retry. I'll use the card selector from the element registry to ensure I'm clicking the correct target.

18. Validation Rules and Evidence Capture

This section defines the validation rules the agent must apply at every step of the demo flow. Each validation type specifies the rule, a concrete example from the loan agent app, and what evidence must be captured to confirm success.

Validation Type	Rule	Example	Evidence to Capture
Page validation	Current route must match expected route and page heading must be visible before any action is taken.	Route /overview and heading 'Overview' visible before reading KPI strip. Route /hil-review and heading 'HIL Review' visible before looking for kanban cards.	Screenshot plus detected route text and page heading text
Element validation	Target element must be visible, enabled, and not covered by another element or modal before clicking.	Taskbar Overview icon visible and clickable before navigating. KRISHNAN DEEPAK kanban card visible in Awaiting Review lane before clicking.	Element ID from Section 6 registry, selector used, bounding box coordinates
Action validation	Expected state change must occur after action is performed. Agent must confirm change before proceeding.	After clicking Approve in HIL modal, modal closes and KRISHNAN DEEPAK card moves to CO Approved lane. After clicking taskbar icon, route changes to expected path.	Before and after screenshots confirming state change
Data validation	Required fields, labels, and values must be visible and readable before narrating or summarising.	All 4 HIL modal metrics (CIBIL, DTI, Bounce, Spending) visible before delivering decision narration. All 8 Application Summary grid fields readable before scrolling timeline.	Captured visible field values, labels confirmed readable
Modal validation	Modal must be fully rendered and correct applicant name and app ID must be confirmed before reading content or clicking action buttons.	ApplicationDetailModal shows KRISHNAN DEEPAK and APP-2026-00012 before scrolling audit trail. ApplicationReviewModal shows correct applicant name before clicking Open page link.	Modal title text, applicant name, app ID all confirmed visible
Timeline validation	Agent must confirm step count before narrating. 8 steps for clean STP cases, 9 steps for escalated or rejected cases. FINAL_DECISION must be the last step read.	MEHTA VARUN timeline has 8 steps ending at FINAL_DECISION with no HIL_ESCALATION. KRISHNAN DEEPAK and PILLAI SAVITA timelines have 9 steps including HIL_ESCALATION before FINAL_DECISION.	Step count confirmed, all step names read, FINAL_DECISION output_data captured as visible text
Narration validation	Agent must narrate only what	CIBIL score, DTI ratio,	Generated narration

Validation Type	Rule	Example	Evidence to Capture
	is visible on screen. No hardcoded values. All metrics read live.	confidence score, risk category all read from screen — not asserted from memory. Queue snapshot counts read live from desktop tiles.	transcript with live-read values clearly indicated
Recovery validation	If any element is not found or action fails, agent must log the failure, attempt one retry, then continue with fallback path from Section 6 if retry fails.	Kanban card not found — refresh /hil-review — retry. Modal does not open — retry click using fallback selector EL-027. Applicant not found in table — use search input EL-020.	Failure event logged, recovery path taken, final outcome recorded

19. Error Handling Matrix

This section defines all error scenarios the agent may encounter during the demo, how to detect them, how to recover, and what to say if the error cannot be resolved.

Error ID	Scenario	Detection	Recovery	Final AI Response If Unresolved
ERR-001	Page fails to load or times out	No page heading visible after 10 seconds. Route does not match expected path.	Refresh the page once. If still failing, navigate back to base_url and retry from desktop.	I was unable to load the expected page after retrying. I will return to the desktop and attempt navigation again.
ERR-002	Target element not found	Selector match fails. Element not visible in expected region. Text search returns no result.	Use fallback coordinates from Section 6 Element Registry. If still not found, scroll the region and retry once.	I could not locate the expected element on this screen. I will use the fallback selector and retry.
ERR-003	Kanban card not found in Awaiting Review lane	KRISHNAN DEEPAK card not visible in first lane after page loads.	Refresh /hil-review once. Check all 4 lanes in order. If card is in Under Review lane, open it from there.	I cannot find the expected kanban card in the Awaiting Review lane. Let me refresh and check all lanes.
ERR-004	HIL modal does not open after card click	No modal overlay visible after clicking kanban card. Page state unchanged.	Retry click using fallback selector EL-027. If modal still does not open after retry, report and skip HIL action.	The modal did not open after clicking the card. I will retry using the fallback selector.
ERR-005	Wrong modal opened	Modal visible but applicant name or app ID does not match expected target.	Close modal using Close button. Re-identify correct card by app ID. Click correct card.	The modal that opened does not match the expected applicant. I will close it and locate the correct card.
ERR-006	ApplicationReviewModal shows no data	Modal overlay visible but audit trail or profile panel is empty or blank.	Close modal. Click Open page link directly from table row to bypass modal and navigate to detail page.	The review modal is not showing data. I will navigate directly to the detail page instead.
ERR-007	HIL action button not visible in modal	Modal footer does not show Approve, Reject, or Modify buttons. Possible read-only modal opened instead of HIL modal.	Close current modal. Return to /hil-review. Ensure card is clicked from kanban board, not from applications table.	The action buttons are not visible. I may have opened the wrong modal. I will return to the HIL kanban board and retry.
ERR-008	Timeline does not scroll or steps not visible	Timeline section present but steps are not rendering or scroll does not respond.	Scroll using keyboard (Page Down). If still unresponsive, read visible steps only and note how many are	The timeline is not fully scrollable. I will read the visible steps and note that some steps could not be confirmed.

Error ID	Scenario	Detection	Recovery	Final AI Response If Unresolved
			visible.	
ERR-009	Wrong page opened from applications table	Route or applicant name on detail page does not match expected applicant.	Click Back to Dashboard. Re-locate correct applicant row using search input EL-020. Click correct Open page link.	The wrong application detail page opened. I will go back and locate the correct applicant.
ERR-010	Unexpected data state — no rows in applications table	Table body is empty or no rows visible after page loads.	Clear all filters using status and risk dropdowns. If table is still empty, refresh /applications once.	No applications are visible in the table. I will clear the filters and refresh to restore the view.
ERR-011	Post-approval — kanban card does not move to CO Approved lane	After clicking Approve, card still visible in Awaiting Review lane after 3 seconds.	Refresh /hil-review. Confirm card is now in CO Approved lane. If not, report that UI update may be delayed.	The card did not move immediately after approval. I will refresh the board to confirm the approval was recorded.
ERR-012	Navigation icon in taskbar not responding	Click on Overview, HIL Review, or Applications icon does not trigger route change.	Retry click once. If no response, use direct URL navigation to the expected route.	The taskbar icon did not respond. I will navigate directly to the expected route.

20. Humanoid Narration Script Patterns

This section defines the narration script patterns the humanoid must follow at every stage of the demo. Each pattern provides a speaker notes template and a concrete example from the loan agent flow.

Situation	Speaker Notes Template	Example
Before navigating to a new page	I am going to navigate to [page] so we can [reason].	I am going to navigate to the HIL Review board so we can see the cases that are awaiting a credit officer decision.
While page is loading	[Page] is loading. Once it opens, I will look for [specific items].	The Overview Dashboard is loading. Once it opens, I will look for the KPI strip, the decision distribution chart, and the 7-step pipeline.
After page opens	The [page] is now visible. I can see [visible regions and items].	The HIL Review board is now visible. I can see four kanban lanes — Awaiting Review, Under Review, CO Approved, and CO Rejected — each with a card count badge.
Before reading metrics or data	Let me read [data area] to understand [reason].	Let me read the key metrics panel on the right side of the HIL modal to understand the financial profile of this applicant before making a decision.
While reading live values	I can see [field name] shows [live read value]. This tells me [interpretation].	I can see the CIBIL Score shows [read value]. This tells me the applicant has a strong credit history. The DTI Ratio shows [read value], which is within the 40% policy limit.
Before scrolling timeline	I will now scroll through the agent timeline step by step. This shows exactly how the system processed this application from start to final decision.	I will now scroll through the agent timeline step by step. This shows exactly how the system processed MEHTA VARUN's application from KYC validation through to the final approval decision.
After reading each timeline step	Step [number] — [step name] — [tool name]. The agent [what happened]. [Key output].	Step 5 — CIBIL Fetch — CIBIL API. The agent retrieved the credit bureau score. CIBIL score is 745 with no adverse flags recorded.
Before making HIL decision	I have reviewed the full audit trail and all four key metrics. Based on [summary of findings], my decision is to [approve/reject/modify].	I have reviewed the full audit trail and all four key metrics. The CIBIL score is strong, the DTI is within policy, bounce is zero, and the income mismatch is within tolerance. My decision is to approve this application.
When explaining a rejection case	This application was rejected because [primary reason]. The key indicator was [metric] which [what it showed].	This application was rejected because the DTI ratio exceeded the policy limit. The key indicator was a DTI of 73.67% against a hard limit of 40%, which signals the applicant cannot service the requested loan given current

Situation	Speaker Notes Template	Example
		obligations.
When uncertain about a visible value	I can see [known element], but [uncertain element] is not clearly readable. I will [safe next step].	I can see the timeline step heading, but the output data block text is small. I will zoom into this region before narrating the values.
When recovering from an error	The expected [element/page] is not visible. I will [recovery action] and retry.	The expected kanban card is not visible in the Awaiting Review lane. I will refresh the HIL Review board and check all lanes before retrying.
Final summary narration	We have now walked through [screens covered]. The key takeaway is [main insight]. The credit officer retains final authority over every decision — the agent recommends, the human decides.	We have now walked through the full loan agent demo — the overview dashboard, the HIL review and approval for KRISHNAN DEEPAK, and the application detail views for all three cases. The key takeaway is that the system automates data extraction, risk scoring, and preliminary recommendations, but the credit officer retains final authority over every decision — the agent recommends, the human decides.

21. Copy-and-Fill Blank Action Matrix

This matrix documents the complete 28-action demo flow in execution order. Every action maps directly to the element registry (Section 6), screen inventory (Section 4), and action examples (Section 8).

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
1	ACT-001	SCR-001-DESKTOP	Browser navigation	EL-001	base_url	base_url runtime parameter	Desktop home page loads with queue snapshot, hero card, taskbar visible	Background visible, hero card visible, taskbar visible	I am navigating to the loan agent dashboard to begin the demo.	Retry navigation once. If still failing, check base_url parameter.
2	ACT-002	SCR-001-DESKTOP	Explain screen	EL-002, EL-003, EL-004	Queue Snapshot (20,30,380,280), Decision Mix (1060,30,1410,280), Hero card (310,260,1130,520)	None	No UI change — narration only	All 4 queue tiles readable, all 3 decision mix bars readable, hero card text visible	The desktop has loaded. I can see today's queue snapshot on the left, the decision mix on the right, and the AI Loan Agent hero card in the centre. Let me read the live values before navigating.	If any tile is unreadable, note it and continue with visible values.
3	ACT-003	SCR-001-DESKTOP	Click	EL-005	a[aria-label='Overview'] (~676,835)	None	Route changes to /overview, Overview Dashboard heading visible	Route = /overview, KPI strip visible	I am navigating to the Overview Dashboard to review the full system metrics and distributions.	If icon unresponsive, navigate directly to base_url/overview.
4	ACT-004	SCR-002-OVERVIEW	Read KPI strip	EL-008	REG-KPI-STRIP (0,0,1440,90)	None	No UI change — narration only	All 5 KPI cards readable with values	Let me read the five KPI cards: [read Applications Today], [read STP Approved %], [read HIL	If any KPI is unreadable, note it and continue.

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
									Queue], [read Rejected], [read Avg TAT].	
5	ACT-005	SCR-002-OVERVIEW	Explain screen	EL-009, EL-010, EL-011, EL-012, EL-013	REG-CHARTS (0,90,1440,540)	None	No UI change — narration only	All 5 charts visible with labels	I can see five charts: Decision Distribution, Risk Distribution, Employer Split, Daily Batch Volume, and Top Rejection Reasons. Let me explain each.	If chart is blank, note it and continue with visible charts.
6	ACT-006	SCR-002-OVERVIEW	Explain screen	EL-014	REG-PIPELINE (0,540,1440,640)	None	No UI change — narration only	All 7 pipeline steps visible and labeled	This is the 7-step AI pipeline. Let me walk through each stage: Data Intake, Feature Engineering, Rule Engine, LLM Reasoning, HIL Review, Audit Trail, LMS Push.	If pipeline strip is not visible, scroll down once.
7	ACT-007	SCR-002-OVERVIEW	Explain screen	EL-015, EL-016, EL-017, EL-018	REG-METRICS (0,640,1440,820)	None	No UI change — narration only	All 4 metric cards readable	Let me read the four performance cards: [read STP Rate], [read Decision Accuracy], [read False Reject Rate], [read LMS Push Success].	If any metric is unreadable, note it and continue.
8	ACT-008	SCR-002-OVERVIEW	Click	EL-006	a[aria-label='HIL Review'] (~720,83	None	Route changes to /hil-review, kanban	Route = /hil-review, gate card and	I am now navigating to the HIL Review board to see the	If icon unresponsive, navigate directly

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
					5)		board visible	4 lanes visible	cases awaiting credit officer decision.	to base_url /hil-review.
9	ACT-009	SCR-005-HIL-REVIEW	Explain screen	EL-019	REG-HIL-LANES (20,160, 1410,850)	None	No UI change — narration only	All 4 lane headers visible with count badges	I can see the HIL kanban board with four lanes. Let me read the lane headers and counts: [read Awaiting Review count], [read Under Review count], [read CO Approved count], [read CO Rejected count].	If count badges are missing, read lane headers only.
10	ACT-010	SCR-005-HIL-REVIEW	Click	EL-027	div.cursor-pointer.rounded-lg in Awaiting Review lane containing KRISHNAN DEEPAK	target_applicant_2 = KRISHNAN DEEPAK	ApplicationDetailModal opens as overlay	Modal title 'Application Review & Status' visible, KRISHNAN DEEPAK name and APP-2026-00012 confirmed	I can see KRISHNAN DEEPAK's application in the Awaiting Review lane. Let me click on it to open the HIL review modal.	If card not found, refresh /hil-review once. Check all lanes. Use fallback: locate card by app ID APP-2026-00012.
11	ACT-011	SCR-006-HIL-MODAL	Validate modal	EL-028	Modal header (30,10,1200,100)	None	No UI change — validation only	Modal title visible, KRISHNAN DEEPAK and APP-2026-00012 confirmed	The HIL modal has opened. I can confirm this is KRISHNAN DEEPAK's application — APP-2026-00012. Let me now review the audit trail.	If wrong applicant shown, close modal and re-click correct card.
1	ACT-	SCR-006-	Scroll	EL-034	section.	None	Audit	All audit	Let me scroll	If scroll

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
2	012	HIL-MODAL			max-h-[400px]. overflow-y-auto, scroll_increment_px: 120		events scroll into view	events narrated: KYC, Document Analysis, Income Verification, Bank Analysis, CIBIL Fetch, Feature Engineering, Decision Engine, HIL Escalation	through the full audit trail. [Read each event in sequence]. I can see the complete processing history for this application.	does not respond, use keyboard Page Down. Read visible events only if scroll is blocked.
13	ACT-013	SCR-006-HIL-MODAL	Read data	EL-035, EL-036, EL-037, EL-038	Key Metrics grid (480,410,1200,800)	None	No UI change — narration only	All 4 metrics readable: CIBIL, DTI, Bounce Frequency, Spending Ratio	Now let me read the four key metrics. [Read CIBIL Score]. [Read DTI Ratio and badge]. [Read Bounce Frequency]. [Read Spending Ratio]. These metrics drive my decision.	If any metric cell is unreadable, note the cell label and continue.
14	ACT-014	SCR-006-HIL-MODAL	Explain screen	EL-028	Full modal (30,10,1200,880)	None	No UI change — narration only	Narration delivered covering metrics, risk, and reasoning	I have reviewed the full audit trail and all four metrics. CIBIL is 789 — strong credit profile. DTI is 19.45% — well within the 40% policy limit. Bounce is zero, spending ratio is moderate. The income mismatch	None — this is a narration-only step.

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
									was flagged but is within tolerance. My assessment is that this application meets the criteria for approval.	
15	ACT-015	SCR-006-HIL-MODAL	Click	EL-041	footer button.bg-emerald-600 (~1285,843)	hil_action = approve	Modal closes, KRISHNAN DEEPAK card moves to CO Approved lane	Modal closed, card visible in CO Approved lane within 3 seconds	I am clicking Approve. This records my decision and moves the application to the CO Approved lane.	If button not visible, confirm correct modal is open (HIL modal, not review modal). Retry click once.
16	ACT-016	SCR-006-HIL-MODAL	Explain screen	EL-019	CO Approved lane (~720,160,1060,850)	None	No UI change — narration only	Narration confirms approval and transitions to next step	The approval has been recorded. I can see KRISHNAN DEEPAK's card has moved to the CO Approved lane. The credit officer decision is final and has been logged in the audit trail. Now let me navigate to the applications register.	If card has not moved, refresh / hil-review and confirm card is in CO Approved lane.
17	ACT-017	SCR-006-HIL-MODAL	Click	EL-007	a[aria-label='Applications'] (~764,835)	None	Route changes to /applications, applications table visible	Route = /applications, table visible with rows	I am navigating to the applications register to show the full batch.	If icon unresponsive, navigate directly to base_url /applications.

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
18	ACT-018	SCR-003-APPLICATIONS	Explain screen	EL-020, EL-021	Filter bar (600,100,1420,165), Mini-charts (0,220,1440,380), Table header (0,380,1440,420)	None	No UI change — narration only	Filter bar, mini-charts, and table header all visible	Here is the full applications register. I can see the search bar, status and risk filters, the distribution mini-charts above, and the full application table below. Let me now open the first case — the clean auto-approval.	If table is empty, clear all filters and refresh once.
19	ACT-019	SCR-003-APPLICATIONS	Click	EL-025	tr containing MEHTA VARUN, Open page link in Route column	target_applicant_1 = MEHTA VARUN	Route changes to /applications/APP-2026-00001, detail page loads	Route = /applications/APP-2026-00001, MEHTA VARUN heading visible	I am opening MEHTA VARUN's application — APP-2026-00001. This is the clean straight-through approval case.	If Open page link not found, use search input EL-020 to filter by MEHTA VARUN, then click link.
20	ACT-020	SCR-004-APP-DETAIL	Read data	EL-039	Application Summary grid (30,200,1200,340)	None	No UI change — narration only	All 8 summary fields readable	Let me read the application summary. [Read Employer], [Read Declared Income], [Read Verified Income — confirm match], [Read CIBIL 745], [Read DTI 23.91%], [Read Risk Category — Low], [Read	If any field is unreadable, note it and continue.

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
									Decision — Conditional Approval], [Read Confidence Score — 100%].	
21	ACT-021	SCR-004-APP-DETAIL	Scroll	EL-040	REG-TIMELINE (30,400, 1200,1000+), scroll through all 8 steps	None	All 8 timeline steps scroll into view	8 steps confirmed, FINAL_DECISION visible at bottom, outcome read as visible text	Let me scroll through the full agent timeline for MEHTA VARUN. [Read each of 8 steps: KYC, Document Analysis, Income Verification, Bank Analysis, CIBIL Fetch, Feature Engineering, Decision Engine, FINAL_DECISION]. This is a clean STP case — no HIL escalation step.	If timeline does not scroll, use keyboard Page Down. Read visible steps only if scroll is blocked.
22	ACT-022	SCR-004-APP-DETAIL	Click	EL-043	a:has-text('Back to Dashboard') (~1200-1400, 80-120)	None	Route changes to /applications, table reloads	Route = /applications, table visible	I am returning to the applications table to open the next case.	If back link not found, navigate directly to base_url /applications.
23	ACT-023	SCR-003-APPLICATIONS	Click	EL-026	tr containing KRISHNAN DEEPAK, Open page link in Route column	target_applicant_2 = KRISHNAN DEEPAK	Route changes to /applications/APP-2026-00012, detail page loads	Route = /applications/APP-2026-00012, KRISHNAN DEEPAK heading visible	Now let me open KRISHNAN DEEPAK's application — APP-2026-00012. This is the case I just approved through the HIL review.	If Open page link not found, use search input EL-020 to filter by KRISHNAN DEEPAK,

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
										then click link.
24	ACT-024	SCR-004-APP-DETAIL	Scroll	EL-040	REG-TIMELINE (30,400, 1200,1000+), scroll through all 9 steps	None	All 9 timeline steps scroll into view including HIL_ESCALATION	9 steps confirmed, HIL_ESCALATION step visible with trigger_reason block, FINAL_DECISION visible at bottom	Let me scroll through KRISHNAN DEEPAK's timeline. This case has 9 steps — the extra step is HIL Escalation which explains why it was routed to me for review. [Read each step including HIL_ESCALATION trigger_reason]. [Read FINAL_DECISION outcome as visible].	If HIL_ESCALATION step is not visible, scroll further. Note step count if different from expected 9.
25	ACT-025	SCR-004-APP-DETAIL	Click	EL-043	a:has-text('Back to Dashboard') (~1200-1400, 80-120)	None	Route changes to /applications, table reloads	Route = /applications, table visible	I am returning to the applications table to open the final case.	If back link not found, navigate directly to base_url /applications.
26	ACT-026	SCR-003-APPLICATIONS	Click	EL-024	tr containing PILLAI SAVITA, Open page link in Route column	target_applicant_3 = PILLAI SAVITA	Route changes to /applications/APP-2026-00003, detail page loads	Route = /applications/APP-2026-00003, PILLAI SAVITA heading visible	Finally let me open PILLAI SAVITA's application — APP-2026-00003. This is the hard rejection case.	If Open page link not found, use search input EL-020 to filter by PILLAI SAVITA, then click link.
27	ACT-027	SCR-004-APP-DETAIL	Scroll	EL-040	REG-TIMELINE (30,400, 1200,10	None	All 9 timeline steps scroll into view	9 steps confirmed, FINAL_DECISION	Let me scroll through PILLAI SAVITA's timeline.	If timeline does not scroll, use

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
					00+), scroll through all 9 steps			visible at bottom showing rejection, DTI breach visible in Feature Engineering step	[Read each step]. The key step to note is Feature Engineering — this is where the DTI breach was detected. [Read DTI 73.67% against 40% policy limit]. [Read FINAL_DECISION outcome as visible].	keyboard Page Down. Read visible steps only.
28	ACT-028	SCR-004-APP-DETAIL	Explain screen	EL-040	Full detail page	None	No UI change — narration only	Closing narration delivered	We have now walked through all three cases — MEHTA VARUN as a clean auto-approval, KRISHNAN DEEPAK as a human-reviewed approval, and PILLAI SAVITA as a hard rejection due to DTI breach. The key takeaway: the system automates extraction, scoring, and recommendation — but the credit officer retains final authority over every decision. The agent recommends. The human	None — this is the closing narration step.

Seq	Action ID	Screen ID	Action Type	Target Element ID	Location / Selector	Required Input	Expected Response	Validation	Speaker Notes	Fallback
									decides.	

22. Copy-and-Fill Blank Page Definition

This section provides a filled page definition for each of the 6 screens in the demo flow. Each table follows the standard page definition format and can be used as a reference for scripting new screens.

SCR-001-DESKTOP — Desktop Home

Field	Value to Fill
screen_id	SCR-001-DESKTOP
screen_name	Loan Agent Desktop Home
url_or_route	/
business_purpose	Entry point for the demo. Shows today's queue snapshot, decision mix, and system hero card. Agent reads live queue data and navigates to other screens via the taskbar.
entry_condition	base_url loaded in browser. Desktop background visible.
exit_condition	Agent clicks one of the three taskbar icons (Overview, HIL Review, Applications) to navigate away.
layout_regions	Top-left: Queue Snapshot panel (4 metric tiles). Top-right: Decision Mix panel (3 progress bars). Centre: Hero card (macOS chrome, AI Loan Agent title). Bottom: Taskbar dock (3 icons + clock).
primary_elements	EL-002 (Need Review tile), EL-003 (Reviewed tile), EL-004 (Accepted tile), EL-008 (Rejected tile), EL-005 (Overview icon), EL-006 (HIL Review icon), EL-007 (Applications icon)
secondary_elements	Hero card subtitle, clock display, Decision Mix progress bars
popups_expected	None
artifacts_available	None
validation_rules	Hero card visible with 'AI Loan Agent' heading. All 4 queue snapshot tiles readable. All 3 decision mix bars readable. All 3 taskbar icons visible.
speaker_notes_before	The loan agent desktop has loaded. I can see today's queue snapshot on the left with live counts, and the decision mix on the right showing the ratio of accepted, rejected, and under-review cases.
speaker_notes_after	I have reviewed the desktop overview. Let me now navigate to the dashboard for the full KPI breakdown.
fallback_navigation	If desktop fails to load, refresh base_url once. If taskbar icons are missing, navigate directly via URL.

SCR-002-OVERVIEW — Overview Dashboard

Field	Value to Fill
screen_id	SCR-002-OVERVIEW

Field	Value to Fill
screen_name	Overview Dashboard
url_or_route	/overview
business_purpose	Shows the full system performance picture: 5 KPI cards, 3 doughnut charts, 2 bar charts, 7-step pipeline strip, and 4 bottom metric cards. Agent narrates all sections before moving to HIL review.
entry_condition	Agent clicks Overview icon from taskbar. Route = /overview. Page heading visible.
exit_condition	Agent clicks HIL Review icon from taskbar after completing narration.
layout_regions	Top: KPI strip (5 cards). Middle: 3 doughnut charts (Decision, Risk, Employer) + 2 bar charts (Volume, Rejection Reasons). Below charts: 7-step pipeline strip. Bottom: 4 metric cards (STP Rate, Accuracy, False Reject, LMS Push).
primary_elements	EL-008 (KPI strip), EL-009 (Decision Distribution chart), EL-010 (Risk Distribution chart), EL-011 (Employer Split chart), EL-012 (Daily Volume chart), EL-013 (Rejection Reasons chart), EL-014 (Pipeline strip), EL-015 to EL-018 (metric cards)
secondary_elements	Chart legends, pipeline step labels, metric card trend indicators
popups_expected	None
artifacts_available	None
validation_rules	Route = /overview. All 5 KPI cards readable. All 5 charts visible with labels. Pipeline strip shows all 7 steps. All 4 metric cards readable.
speaker_notes_before	I am now on the Overview Dashboard. Let me read the KPI strip first, then walk through the charts, explain the pipeline, and finish with the performance metrics.
speaker_notes_after	I have reviewed the full overview. The system metrics and distributions are clear. Let me now navigate to the HIL Review board to handle the pending cases.
fallback_navigation	If page does not load, navigate directly to base_url/overview.

SCR-005-HIL-REVIEW — HIL Review Kanban

Field	Value to Fill
screen_id	SCR-005-HIL-REVIEW
screen_name	HIL Review Kanban Board
url_or_route	/hil-review
business_purpose	Shows all applications requiring human credit officer review in a 4-lane kanban board. Agent reads lane counts, locates KRISHNAN DEEPAK card in Awaiting Review lane, and clicks to open HIL modal.
entry_condition	Agent clicks HIL Review icon from taskbar. Route = /hil-review. Gate card and 4 lanes visible.
exit_condition	Agent clicks KRISHNAN DEEPAK kanban card and ApplicationDetailModal opens.
layout_regions	Top: Gate card header (title + subtitle). Below: 4 kanban lanes side by side — Awaiting Review, Under Review, CO Approved, CO Rejected. Each lane has header with count badge and stacked kanban cards below.
primary_elements	EL-019 (lane headers + count badges), EL-027 (KRISHNAN DEEPAK kanban card in Awaiting Review lane)

Field	Value to Fill
secondary_elements	Other kanban cards in all lanes, gate card subtitle text
popups_expected	ApplicationDetailModal opens when kanban card is clicked
artifacts_available	None
validation_rules	Route = /hil-review. Gate card visible. All 4 lane headers visible with count badges. KRISHNAN DEEPAK card visible in Awaiting Review lane.
speaker_notes_before	I am on the HIL Review kanban board. Let me read all four lanes and their counts before opening the pending case.
speaker_notes_after	I have identified KRISHNAN DEEPAK's application in the Awaiting Review lane. Let me open it to review the details.
fallback_navigation	If KRISHNAN DEEPAK card not found in Awaiting Review, refresh page once. Check all 4 lanes. Use app ID APP-2026-00012 to identify correct card.

SCR-006-HIL-MODAL — HIL Application Modal

Field	Value to Fill
screen_id	SCR-006-HIL-MODAL
screen_name	HIL Application Detail Modal (ApplicationDetailModal)
url_or_route	Modal overlay on /hil-review
business_purpose	Allows credit officer to review the full audit trail, applicant profile, and key metrics before making a final decision. Three decision actions available: Approve, Reject, Modify and Approve.
entry_condition	Agent clicks kanban card in Awaiting Review lane. Modal overlay appears on /hil-review.
exit_condition	Agent clicks one of three decision buttons (Approve, Reject, Modify and Approve). Modal closes and card moves to appropriate lane.
layout_regions	Header: Modal title + Close button. Left panel: Scrollable audit timeline (section.max-h-[400px].overflow-y-auto). Right panel top: Applicant name, app ID, status badge, risk score bar. Right panel bottom: 4-cell key metrics grid (CIBIL, DTI, Bounce, Spending). Footer: 4 action buttons.
primary_elements	EL-028 (modal header), EL-034 (audit trail scroll section), EL-035 (CIBIL metric), EL-036 (DTI metric), EL-037 (Bounce metric), EL-038 (Spending metric), EL-041 (Approve button), EL-042 (Reject button), EL-040 (Modify & Approve button)
secondary_elements	Risk score progress bar, status badge, Close button (white), Cancel button
popups_expected	None within modal — modal itself is the overlay
artifacts_available	None
validation_rules	Modal title 'Application Review & Status' visible. KRISHNAN DEEPAK and APP-2026-00012 confirmed. All 4 metric cells readable. All 3 decision buttons visible with correct colors (green/red/amber).
speaker_notes_before	The HIL modal has opened for KRISHNAN DEEPAK — APP-2026-00012. Let me first scroll through the full audit trail on the left, then read all four key metrics before making my decision.
speaker_notes_after	I have reviewed all evidence. Based on the CIBIL score, DTI ratio, and supporting data, I am

Field	Value to Fill
	confident in my decision.
fallback_navigation	If modal does not open, retry card click using fallback selector EL-027. If action buttons are missing, this may be ApplicationReviewModal (read-only) — close and re-click from HIL kanban, not applications table.

SCR-003-APPLICATIONS — Applications Table

Field	Value to Fill
screen_id	SCR-003-APPLICATIONS
screen_name	Applications Register Table
url_or_route	/applications
business_purpose	Shows the full batch of processed applications in a searchable, filterable table. Agent uses this screen to locate and open individual application detail pages for MEHTA VARUN, KRISHNAN DEEPAK, and PILLAI SAVITA.
entry_condition	Agent clicks Applications icon from taskbar OR navigates from post-approval HIL screen. Route = /applications. Table visible with rows.
exit_condition	Agent clicks Open page link for a target applicant. Route changes to /applications/:id.
layout_regions	Top: Filter bar (search input, status dropdown, risk dropdown, export button). Middle: 3 mini-charts (Decision, Risk, Confidence distributions). Bottom: Application table with header row and data rows.
primary_elements	EL-020 (search input), EL-021 (status filter), EL-022 (risk filter), EL-023 (export button), EL-024 (PILLAI SAVITA Open page link), EL-025 (MEHTA VARUN Open page link), EL-026 (KRISHNAN DEEPAK Open page link)
secondary_elements	3 mini-charts, table header row, row count badge
popups_expected	ApplicationReviewModal opens when any table row body is clicked (read-only). Clicking Open page link bypasses modal and navigates directly to detail page.
artifacts_available	None
validation_rules	Route = /applications. Table visible with rows. Search input visible. All 3 target applicant rows locatable. Open page links visible in Route column.
speaker_notes_before	Here is the full applications register. Let me note the distribution charts above the table, then locate each of our three target applicants.
speaker_notes_after	I have opened the target application detail page. Let me read the summary and timeline.
fallback_navigation	If target applicant row not found, use search input EL-020 to filter by name or app ID. If table is empty, clear all filters and refresh.

SCR-004-APP-DETAIL — Application Detail Page

Field	Value to Fill
screen_id	SCR-004-APP-DETAIL

Field	Value to Fill
screen_name	Application Detail Page
url_or_route	/applications/:id (APP-2026-00001, APP-2026-00003, APP-2026-00012)
business_purpose	Shows the full application record for a single applicant — 8-field summary grid and complete agent timeline. Agent reads the summary grid then scrolls and narrates every timeline step through to FINAL_DECISION.
entry_condition	Agent clicks Open page link from applications table. Route = /applications/:id. Applicant name heading visible.
exit_condition	Agent clicks Back to Dashboard link after completing timeline narration. Route returns to /applications.
layout_regions	Header: Applicant name (large), app ID, Back to Dashboard link. Below header: Application Summary grid (8 fields, 2 columns). Below summary: Agent Timeline section header + subtitle. Below header: Scrollable timeline steps (data-driven count: 8 for STP cases, 9 for escalated/rejected cases).
primary_elements	EL-039 (Application Summary grid), EL-040 (Timeline steps scroll region), EL-043 (Back to Dashboard link)
secondary_elements	Applicant name heading, app ID label, timeline section header and subtitle
popups_expected	None
artifacts_available	None
validation_rules	Route matches expected /applications/:id. Applicant name heading matches target applicant. All 8 summary fields readable. Timeline section visible. FINAL_DECISION step present at bottom of timeline. Step count: 8 for MEHTA VARUN, 9 for KRISHNAN DEEPAK and PILLAI SAVITA.
speaker_notes_before	I am on the application detail page for [applicant name]. Let me first read all eight fields in the summary grid, then scroll through the complete agent timeline step by step.
speaker_notes_after	I have completed the full timeline walkthrough. The FINAL_DECISION step shows [read visible outcome]. Let me return to the applications table.
fallback_navigation	If wrong detail page loads, click Back to Dashboard and re-locate correct applicant row. If Back link not visible, navigate directly to base_url/applications.

23. Master JSON Example for AI Runtime

This is the complete machine-readable execution configuration for the Loan Underwriting Agent demo. The AI runtime uses this JSON to initialise parameters, apply runtime rules, and validate completion criteria at the end of the flow.


```
{
  "flow_id": "FLOW-LOAN-UNDERWRITING-DEMO-001",
  "prototype": "AI Loan Underwriting Agent Dashboard",
  "entry_screen_id": "SCR-001-DESKTOP",
  "parameters": {
    "base_url": "http://localhost:3000",
    "viewport_width": 1440,
    "viewport_height": 900,
    "session_id": "DEMO-SESSION-001",
    "target_applicant_1_name": "MEHTA VARUN",
    "target_applicant_1_id": "APP-2026-00001",
    "target_applicant_1_role": "happy_path_stp_approval",
    "target_applicant_2_name": "KRISHNAN DEEPAK",
    "target_applicant_2_id": "APP-2026-00012",
    "target_applicant_2_role": "hil_escalation_approval",
    "target_applicant_3_name": "PILLAI SAVITA",
    "target_applicant_3_id": "APP-2026-00003",
    "target_applicant_3_role": "hard_rejection_dti_breach",
    "hil_action": "approve",
    "hil_target_applicant": "KRISHNAN DEEPAK",
    "hil_target_app_id": "APP-2026-00012"
  },
  "runtime_rules": {
    "preferred_locator_order": ["aria-label","data-testid","css-selector","visible
text","fallback coordinates"],
    "wait_timeout_seconds": 10,
    "screenshot_policy": "capture_on_major_screen_change",
    "narration_policy": "speak_before_and_after_each_major_action",
    "do_not_infer_unreadable_content": true,
    "do_not_hardcode_metric_values": true,
    "read_all_values_live_from_screen": true,
    "timeline_step_count_policy": "data_driven_do_not_assert_exact_count",
    "final_decision_label_policy": "read_visible_text_do_not_assert_exact_label",
    "modal_action_policy": "match_button_by_color_before_clicking",
    "retry_policy": "retry_once_on_element_not_found_then_use_fallback",
    "hil_modal_source": "kanban_card_only_not_applications_table"
  },
  "screens": [
    { "screen_id": "SCR-001-DESKTOP",      "route": "/",               "entry_action": "ACT-
001" },
    { "screen_id": "SCR-002-OVERVIEW",     "route": "/overview",       "entry_action": "ACT-
003" },
    { "screen_id": "SCR-005-HIL-REVIEW",   "route": "/hil-review",     "entry_action": "ACT-
008" },
    { "screen_id": "SCR-006-HIL-MODAL",    "route": "modal/hil-review","entry_action": "ACT-
010" },
    { "screen_id": "SCR-003-APPLICATIONS", "route": "/applications",   "entry_action": "ACT-
017" },
    { "screen_id": "SCR-004-APP-DETAIL",   "route": "/applications/:id","entry_action": "ACT-
019" }
  ],
  "action_sequence": [
    "ACT-001","ACT-002","ACT-003","ACT-004","ACT-005","ACT-006","ACT-007",
    "ACT-008","ACT-009","ACT-010","ACT-011","ACT-012","ACT-013","ACT-014",
    "ACT-015","ACT-016","ACT-017","ACT-018","ACT-019","ACT-020","ACT-021",
    "ACT-022","ACT-023","ACT-024","ACT-025","ACT-026","ACT-027","ACT-028"
  ],
  "completion_criteria": [
    "Desktop queue snapshot and decision mix read and narrated",
```

```
    "Overview dashboard KPIs, charts, pipeline, and metrics narrated",
    "HIL review kanban board lanes read and counts narrated",
    "KRISHNAN DEEPAK HIL modal opened, audit trail scrolled, all 4 metrics read",
    "Approve decision recorded, card moved to CO Approved lane confirmed",
    "MEHTA VARUN detail page opened, summary read, all 8 timeline steps narrated",
    "KRISHNAN DEEPAK detail page opened, 9 steps incl HIL Escalation narrated",
    "PILLAI SAVITA detail page opened, DTI breach identified, all steps narrated",
    "FINAL_DECISION step read for all three applicants",
    "Closing narration delivered - agent recommends, credit officer decides"
  ]
}
```

24. Final Response Object Example

This is the structured response object the agent produces after completing the full demo flow. It records flow status, screens visited, HIL decisions made, applicants reviewed, completion criteria met, and the final spoken summary.


```
{
  "flow_id": "FLOW-LOAN-UNDERWRITING-DEMO-001",
  "status": "completed",
  "screens_visited": [
    "SCR-001-DESKTOP",
    "SCR-002-OVERVIEW",
    "SCR-005-HIL-REVIEW",
    "SCR-006-HIL-MODAL",
    "SCR-003-APPLICATIONS",
    "SCR-004-APP-DETAIL"
  ],
  "actions_executed": 28,
  "hil_decisions": [
    {
      "applicant": "KRISHNAN DEEPAK",
      "app_id": "APP-2026-00012",
      "action": "approve",
      "cibil_read": 789,
      "dti_read": "19.45%",
      "bounce_read": 0,
      "spending_read": 0.53,
      "result": "card_moved_to_co_approved"
    }
  ],
  "applicants_reviewed": [
    {
      "name": "MEHTA VARUN",
      "app_id": "APP-2026-00001",
      "role": "happy_path_stp_approval",
      "timeline_steps": 8,
      "final_decision": "[read from screen]",
      "result": "reviewed"
    },
    {
      "name": "KRISHNAN DEEPAK",
      "app_id": "APP-2026-00012",
      "role": "hil_escalation_approval",
      "timeline_steps": 9,
      "hil_escalation_present": true,
      "final_decision": "[read from screen]",
      "result": "reviewed"
    },
    {
      "name": "PILLAI SAVITA",
      "app_id": "APP-2026-00003",
      "role": "hard_rejection_dti_breach",
      "timeline_steps": 9,
      "dti_breach_confirmed": true,
      "final_decision": "[read from screen]",
      "result": "reviewed"
    }
  ],
  "completion_criteria_met": [
    "Desktop queue snapshot and decision mix read and narrated",
    "Overview dashboard KPIs, charts, pipeline, and metrics narrated",
    "HIL review kanban board lanes read and counts narrated",
    "KRISHNAN DEEPAK HIL modal opened, audit trail scrolled, all 4 metrics read",
    "Approve decision recorded, card moved to CO Approved lane confirmed",
    "MEHTA VARUN detail page opened, summary read, all 8 timeline steps narrated",
```

"KRISHNAN DEEPAK detail page opened, 9 steps incl HIL Escalation narrated",
"PILLAI SAVITA detail page opened, DTI breach identified, all steps narrated",
"FINAL_DECISION step read for all three applicants",
"Closing narration delivered"
],
"humanoid_final_spoken_summary": "We have now completed the full loan agent demo. I reviewed the overview dashboard showing the full system metrics and distributions. I then opened the HIL review board and approved KRISHNAN DEEPAK's application after reviewing the audit trail and confirming the CIBIL score, DTI ratio, bounce frequency, and spending ratio. I then walked through the application detail pages for all three cases — MEHTA VARUN as a clean auto-approval, KRISHNAN DEEPAK as a human-reviewed approval, and PILLAI SAVITA as a hard rejection due to a DTI breach of 73.67% against the 40% policy limit. The key takeaway: the system automates extraction, scoring, and recommendation — but the credit officer retains final authority over every decision. The agent recommends. The human decides.",
"recommended_next_action": "Return to HIL review board to process remaining cases in Awaiting Review lane."
}